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SECTION 1

GENERAL INFORMATION

1.1 PURPOSE

This handbook provides technical advice to USPS managers, supervisors, and elevator mechanics. It is not intended to replace manufacturers' technical and maintenance publications or the *American National Standard Safety Code* (ASME A17.1). However, it will serve as a reference for postal employees to better understand these publications and identify specific procedures and maintenance requirements for postal facilities. Current postal policy published in Supply Management's Maintenance Logistics Bulletin (MLB-CO-14-001 which is available at the Supply Management MLB webpage) requires all USPS locations to use approved vendors for all elevator and escalator maintenance services that exceed the local maintenance staff's capabilities. Since the National Center for Employee Development (NCED) no longer conducts elevator maintenance training or certification classes, there are very few (if any) sites capable of elevator maintenance without vendor support. Always ensure a complete Article 32 analysis is performed when necessary.

1.2 SAFETY

1.2.1 General

This subchapter contains technical and administrative guidance for personnel concerned with the operation, maintenance, and repair of elevators, escalators, and dumbwaiters (vertical transportation equipment). The purpose of these guides is to assure the safety of all persons.

WARNING

Procedures within this handbook may expose employees to hazardous voltages. Before performing these types of procedures employees must don Electrical Work Plan (EWP) Personal Protective Equipment (PPE) in accordance with the current EWP MMO. Failure to comply may cause injury or death.

1.2.2 Responsibilities

1.2.2.1 Operator Personnel

Operators are assigned for car switch operation or non-automatic freight elevators. They must always be certain the car is at the landing before stepping through the doorway. Never assume car is where it was left. Someone else may have moved it.

Operators should wait until doorway is clear of passengers or trucks before attempting to close the hoistway and car doors. Passengers should not be allowed to enter elevators until the elevator has come to a stop at the landing. They should be cautioned to stay clear of closing doors or gates.

Before starting car, operators should make certain that:

- Passengers or material being handled are in a safe position,
- The hoistway door is closed and locked, and
- The car gate is fully closed.

Operators must never, under any circumstances:

- Permit any material to be placed on top of car or suspended beneath it.
- Use an emergency key to open a hoistway door.
- Attempt to repair, adjust, block open, or otherwise make ineffective the hoistway door interlocks, car door or gate contacts, or the emergency release.

Emergency exits in either the side or top of car must be closed at all times when the car is in motion.

Operators must not permit overloading or allow passengers on freight elevators.

The operator must never operate a defective elevator. As soon as it is evident that repairs are necessary, take the elevator to the nearest landing and follow this procedure:

1. Place the operating control in the locked position.
2. Place the emergency stop or safety switch in the off position.
3. Turn off the fans (if being used).
4. Turn out the lights.
5. If possible, close or fasten the doors to avoid the chance of an accident.
6. Immediately report the trouble to supervisor.

1.2.2.2 Maintenance Personnel

Maintenance personnel should have a thorough knowledge of the proper use of hand and power tools, and be familiar with and practice correct methods of hand lifting. They should use proper and advantageous rigging methods and hoisting equipment. They should have appropriate wearing apparel or special protective clothing and safety equipment when required. Loose clothing, neckties, unbuttoned sleeves, etc., should be avoided. They should know precautions to take when working in the machine room, hoistway, on top of car, in the car, and in the pit.

1.2.3 Elevator Safety Requirements

1.2.3.1 Application

All maintenance supervisors and mechanics must comply with the listed procedures.

1.2.3.2 Codes

All elevator maintenance, repairs, and alterations must be in compliance with the Safety Code for Elevators and Escalators, ASME A17-1, and the National Fire Protection Association Code (NFPA) including NFPA 70 (National Electric Code) and those portions of NFPA code related to fire resistance ratings. All elevator mechanics must be fully aware of these codes, understand the parts applicable to elevators, and understand their responsibility for their own personal safety, the safety of fellow employees, and the safety of the public using elevators they service.

1.2.3.3 Procedures

1.2.3.3.1 Improper Operation

When an elevator operates improperly or in a manner which causes doubt about its safety or reliability, it must be immediately removed from service. The cause of the improper operation is determined and corrected, and the elevator is tested prior to returning it to service.

1.2.3.3.2 Controls

Elevator mechanisms or controls must never be manipulated or temporarily altered to expedite repairs. All alterations, both mechanical and electrical, must be approved by a professional engineer and certified for compliance with ASME A17.1. Any alteration to the controller should be approved by the controller's manufacturer.

1.2.3.3.3 Malfunctions

When an elevator malfunction is reported, the elevator must be taken out of service immediately. The person responding to the call, before troubleshooting the malfunction, must:

1. Make sure that no one is on the elevator,
2. Shut it down,
3. Post *Out of Service* signs in front of hoistway door at each level,

Then:

4. Determine the source of the malfunction,
5. Make the proper repairs and, thoroughly test the elevator prior to returning it to service. Makeshift repairs may not be made.

1.2.3.3.4 Signs

Prior to performing any work, however minor, on an elevator, signs must be placed in front of the hoistway door at each level advising that the elevator is out of service. Signs supported by a chain suspended from each side of the hoistway door, attached to the door facing with magnets, are recommended. Magnetic signs attached directly to the hoistway door may not be used since they would not be visible when the door is open

1.2.3.3.5 Stalled Elevators

If persons are trapped in a stalled elevator, follow the procedures in the Integrated Emergency Management Plan.

1.2.3.4 Tools and Equipment

Elevator mechanics must be provided with the proper tools and equipment for servicing elevators. Only high quality meters may be used to test elevator circuits. Test lamps must never be used since the lamps will often draw sufficient current to operate relays.

1.2.3.5 Machine Room Doors

Elevator machine room doors must be self-closing and self-locking, and be provided with a spring type lock that permits the doors to be opened from the inside without a key. These doors must be kept closed and locked.

1.2.3.6 Personnel Restriction

Only qualified elevator mechanics, inspectors, or persons in their company are allowed to enter elevator machine room or pit. Only a qualified elevator mechanic who is familiar with the equipment may enter a machine room to try to get a stalled elevator started.

1.2.4 Control of Elevators During Emergency

1.2.4.1 Fire

Elevators must not be used for evacuation during a fire emergency because of the danger of entrapment and subsequent injury or death from heat or smoke. In some buildings, elevators are equipped with fireman control (i.e., when a fire alarm is signaled, the elevators will move to the appropriate exit floor; they can then be operated only with a special key by fire department personnel who have equipment for life support and prevention of entrapment).

1.2.4.2 Earthquake

Elevators must not be used during or immediately after an earthquake. They must be stopped at the nearest landing and not returned to service until they have been inspected to assure that they are safe and further damage will not be caused by their operation.

1.2.4.3 Other Emergencies

Control of elevators and direction of their operation must be exercised by the USPS buildings manager or officer in charge, according to USPS directives dealing with the particular type of emergency.

1.2.5 Safe Work Practices

1.2.5.1 Securing Equipment Before Working On It

Elevators, escalators, and dumbwaiters must be taken out of service before any work is started (See Paragraph 1.2.3.3.3). Open the switch, lock out in compliance with local Energy Control Procedure. The tag must remain in place and the circuit must remain open until the person who tagged and locked it removes the tag and lock. If operation of the elevator is required for troubleshooting or testing, the signs specified in Paragraph 1.2.3.3.4 must remain in place until all work is completed.

1.2.5.2 Riding on Roof of Elevator Car

When work on an elevator requires riding on the top of the car, a thoroughly competent operator must move the car. When more than one person is working on the elevator, instructions to move the car are given by one person only, who must ensure that all other persons are in a safe position before the car is moved. The roofs of cars must be kept clean and free of tools and/or spare parts.

CAUTION

Do not hold on to ropes, as this can result in serious injury when 2 to 1 roping is used.

Use car top controls if available, thus negating the need for someone in the car. If more than one person is on top of the car, the car top controls should be operated by only one person. Before moving the car that person must:

1. Check visually to be sure that everyone is clear and ready.
2. Announce in a loud clear voice the direction that the car will be moved, i.e., going up or going down.
3. Get verbal assurance from everyone that they are clear and are ready.

1.2.5.3 Welding or Burning in Hoistway

1.2.5.3.1 Hot Work Permit

Whenever it is necessary to use an electric welding or a burning torch in the hoistway, a "Hot Work Permit" is required in compliance with MS-56 Chapter 5.

1.2.5.3.2 Threat of Fire

Give particular attention to the threat of a fire. A fire extinguisher of the proper type must be readily available. Take extra precaution by using metal shields, welding blankets, etc., to control the action of sparks and drops of molten metal. Workers must be alert for proximity of combustibles, etc., in the vicinity. All elevators in common hoistways must be taken out of service. Be sure the electric welding machine is properly grounded. Torches or furnaces which use gasoline or benzol should not be used in hoistway; use bottled gas burners instead. **NOTE**: The Code (ASME A17.1) prohibits welding repair of parts subject to tension, torsion, or bending on which the support of the elevator car depends.

1.2.5.4 Equipment

1.2.5.4.1 Cleanliness

Clean equipment promotes safety as well as enhancing maintenance and inspection. Do not let oil drippings or excess grease remain on the equipment or in the pit. Use deflectors and/or drip pans and pick up grease from the pit floor with sand or oil absorbing granules. Guide rails that require lubrication must be thoroughly cleaned at least once each year.

1.2.5.4.2 Care of Tools and Portable Items

Keep hand tools and portable equipment in good condition. If tools become dull, worn, or unserviceable, repair or replace them. Be sure all tools, portable equipment, and any loose material is accounted for and removed when work on the equipment is completed.

1.2.5.5 Use of Barricades at Openings

Barricades, portable fences, guards, etc., must be used at all locations where there is a chance that someone might walk or step into an opening while work is in progress.

1.2.5.6 Elimination of Hazardous Protrusions

Give particular attention to access plates and cover (including those over sidewalk elevators, etc.) to ensure that nothing protrudes which would constitute a tripping hazard or which might catch a person's clothes.

1.3 SAFETY INSPECTION AND TESTS

1.3.1 General

Inspections intervals shall be in accordance with the current Management Instruction or Maintenance Logistics Bulletin (MLB).

1.3.1.1 Purpose

The inspection is to assure that the equipment is in safe operating condition and in compliance with the condition of ASME A17.1 in effect when the equipment was installed or modernized. The inspection is primarily for safety. Since most maintenance deficiencies will eventually become safety deficiencies, maintenance deficiencies are often cited.

1.3.1.2 Duties

Inspectors will observe the test and inspect equipment. They will not perform the test and will only intercede if they notice that the test is being performed in a way that is likely to result in personal injury. However, the inspector is not responsible for injury or damage to equipment. The inspector will prepare the inspection reports and certificate. The inspector is usually not an engineer therefore, the inspection report will not necessarily identify the precise work needed to correct the deficiency.

1.3.1.3 Code Requirements

The *American Society Mechanical Engineers for Safety Code for Elevator and Escalators, (ASME A17.1)*, applies. The provisions of this code are constantly changing. Code changes are not retroactive, and elevators are inspected to the code additions in effect on the date of installation or major alteration. However, inspectors sometimes recommend changes to meet current code requirements when they judge the deficiency to be a serious hazard. Also, changes or improvements to meet the current code or local codes can sometimes be made at very little cost and are recommended by the inspector. When major alterations are made to elevators, they shall be upgraded to meet the current code in all respects.

1.3.2 Elevator Tests

1.3.2.1.1 General

Periodic Inspection and Tests lists the tests that are to be witnessed by the elevator inspector. Other tests performed at more frequent intervals by maintenance personnel (such as buffer oil level test) are not listed. Record the date of tests on the Equipment History Record and enter in eMARS as requested by *MS-63, Maintenance Operations.*

1.3.2.2 Test of Safeties

1.3.2.2.1 Purpose

Safeties are tested to assure that they will function properly and stop the car or counterweight evenly and within the range of slide specified in the code. If the slide is too great, the safety may allow the car to go into the pit. If the slide is too short, the rapid deceleration may injure passengers and/or damage the rails, car, rope, etc. **NOTE**: The governor overspeed switch is disabled during these tests and the switch on the safety is temporarily adjusted to trip as late as possible (safety fully applied) for these tests.

1.3.2.2.2 Periodic Test (6 month interval)

This consists of operating the elevator at inspection speed in down direction with no load and tripping the governor by hand. Before performing this test, the safety parts must be carefully inspected to determine that they are in satisfactory condition and the safety jaw - rail clearance is correct. The examination is specified in 8.11.2.2.2.

1.3.2.2.3 Five Year Test

Once every 5 years, after the annual tests have been made, the safeties are tested by operating the car with rated load at rated speed and tripping the governor by hand. The test shall conform with requirements in 8.11.2.

The manufacturer’s specified clearances between the rails and the safety jaws shall be maintained.

1.3.2.3 Governor Tests

1.3.2.3.1 General

The governors are tested to assure that they will function properly. The rope gripping jaws of the governor are actually tested when the safeties are tested, so the test of the governor is primarily concerned with the trip mechanism. The main power supply is opened and locked out during these tests.

1.3.2.3.2 Annual Test

The annual test consists of tripping the governor by hand with the car at rest and carefully examining the mechanism to assure that all parts operate freely without excess looseness and that the trip setting seal is in place. If the seal has been disturbed or there is any other reason to suspect the setting, it should be tested as described in Paragraph 1.3.2.3. The overspeed and slow down switch should be carefully examined.

1.3.2.3.3 Five Year Test

This test consists of removing the governor rope from the governor and rotating the governor wheel by some auxiliary power means (such as a variable speed electric drill with a small rubber wheel in the chuck). As the governor turns, a tachometer is held at the pitch diameter and read when the governor trips. If the trip speed does not conform to the code, it is adjusted and resealed by the inspector. If the tripping cannot be set, the governor should be rebuilt or replaced.

1.3.3 Annual Test of Oil Buffers

The buffer must be tested by fully compressing the plunger. The plunger should then be released and must return to a fully extended position within 90 seconds. Generally, the plunger can be compressed by hand or by standing on it. If it will not compress, a piece of wood or substitute item can be placed between the car or counterweight and the buffer, and then the car or counterweight is run down at inspection speed.

1.3.3.1 Oil Buffer Test (Five Year)

The car oil buffer must be tested with rated load in the car. It is tested by striking the buffer at rated speed.

Before striking the buffer, it will be necessary to jumper out any terminal slowdown switches and any terminal stopping devices.

Where an emergency terminal speed limiting device is provided, test as follows:

1. Determine that the device is operational.
2. With the normal stopping means, and if separate, the normal terminal stopping device deactivated, run the car into the terminal landing with rated load and at operating speed.
3. After the car has been removed from the buffer, note if there has been any excessive leakage of oil.
4. Remove any jumper wires that were installed.

1.3.3.2 Hydraulic Cylinder

Annual leak test of the hydraulic cylinder is required to disclose any small leak in the unexposed portion of the cylinder. Even a pin hole leak is an indication of a weak, corroded cylinder which is likely to fail and cause the car to fall. This test is only required for cylinders that are not exposed to visual inspection. This test is conducted as follows:

1. With no load in the car, locate it at any convenient level, noting and/or marking its exact level.
2. Open the disconnect switch and close manual valve in pit (if installed).
3. Mark exact level of oil in tank and provide for collecting and measuring oil that leaks around the cylinder packing. It is best to conduct this test after the elevator has been out of service for a period so the temperature change of the oil will be negligible.
4. After at least 15 minutes, note the position of the car. A change in position of the car which cannot be accounted for by visible oil leak (around packing) indicates a leak in the cylinder, and operation of the car should not be resumed until repairs are made.

1.3.3.3 Hydraulic Relief Valve Test

1.3.3.3.1 General

All hydraulic elevators require a relief valve between the pump and check valve. The relief valve must be set to open at 125 percent of maximum working pressure and be large enough to relieve the full capacity of the pump without the pressure increasing over 20 percent of the pressure at which the valve first opened.

1.3.3.3.2 Sealing

Relief valves that have exposed means of adjustment must have their means of adjustment sealed by the inspector after being adjusted.

1.3.3.3.3 Working Pressure

Refer to ASME A-17.1. It is the pressure measured at the cylinder of the hydraulic elevator cylinder when lifting car and its rated load at rated speed. This pressure is recorded on a plate permanently mounted on the power unit assembly including the maximum load the elevator is designed to support while loading.

1.3.3.3.4 Method of Test

There are two methods of testing the relief valve setting. The first is as follows:

1. Install an accurate pressure gauge with a mid-scale reading approximately equal to the working pressure of the elevator.
2. Inch the empty car up very slowly until the plunger stop ring engages the top of the cylinder.
3. Continue to apply pressure from the pump and observe the pressure gauge noting the pressure when the relief valve first opens and the pressure when the full output of the pump is relieved.

The gauge is sometimes impossible to read when the valve first relieves because the pressure immediately rises to that required to relieve full capacity. In such cases if the full by-pass pressure is at or below 125% of the working pressure, both conditions have been met and no further tests are needed.

However, if the pressure is greater than 125% but less than 150%, further tests must be conducted to determine that the valve first opened at or below 125%. This is done as follows:

1. Lower the car.
2. Close the main shutoff valve.
3. Operate the pump and slowly open the valve until the pressure gauge reads 125% of working pressure.

If under this condition the car is moving up at greatly reduced speed, it indicates that the valve has opened at or below 125% of working pressure and the test is OK. If not, adjustment or replacement of the valve is required.

1.3.3.4 Flexible Hoses and Fittings

These are permanently marked with the SAE hose type, identification, and required replacement date. These hoses are required to be replaced every 6 years and hydrostatically tested at three times the working pressure every 3 years. This requires removal of the hoses and connecting them to a special pump. As a practical matter, it is often less expensive to replace the hoses every 3 years, and this is usually done.

1.3.3.5 Other Tests

Other tests are required by National Elevator Code for rope hydraulic elevators and stored pressure tanks. However, there are no such units in USPS buildings and they are not covered in this Handbook.

1.3.4 Escalator Test

Test of escalators consists of checking the safety devices for proper operation. This includes:

NOTE

Speed governors are required only where DC drive motors are used.

- Where a speed governor is used, it is manually operated.
- Manually actuate the broken step chain device and observe that it functions properly.
- Manually actuate the broken drive chain device (where chain drive is used) and observe it for proper operation.
- Push emergency stop button with escalator operating in each direction and determine that escalator will not restart without key.
- Check operation of skirt switches where provided.

1.3.5 Periodic Inspection and Tests

Periodic Inspections

Reference Section	Equipment Type	Requirement	Interval in months
8.11.2	Electric Elevators	8.11.2.1	6
8.11.3	Hydraulic Elevators	8.11.3.1	6
8.11.4	Escalators and moving walks	8.11.4.1	6
8.11.5.1	Sidewalk Elevators	8.11.2.1,8.11.3.1	6

Periodic Tests

		Category 1		Category 3		Category 5	
Reference Section	Equipment Type	Requirement	Interval in months	Requirement	Interval in months	Requirement	Interval in months
8.11.2	Electric Elevators	8.6.4.19	12	N/A	N/A	8.6.4.20	60
8.11.3	Hydraulic Elevators	8.6.5.14	12	8.6.5.15	36	8.6.5.16	60
8.11.4	Escalators and moving walks	8.6.8.15	12	N/A	N/A	N/A	N/A
8.11.5.1	Sidewalk Elevators	8.6.7.5.1	12	8.6.7.5.1	36	8.6.7.5.1	60

SECTION 2

EQUIPMENT AND OPERATIONS

2.1 EQUIPMENT

2.1.1 General

The types of equipment found in USPS facilities are listed in this subchapter. Illustrations are included to improve clarity. The definitions are taken from the *American National Standard Safety Code for Elevators, Dumbwaiters, and Escalators*, ASME A17.1. Refer to that publication for a complete definition of all related items

2.1.2 Types of Equipment

2.1.2.1 Dumbwaiters

A hoisting and lowering mechanism with a car of limited capacity and size which moves in guides in a substantially vertical direction and is used exclusively for carrying materials only. Use for carrying any passengers is prohibited.

2.1.2.2 Freight Elevators

An elevator primarily used for carrying freight and on which only the operator and the persons necessary for unloading and loading the freight are permitted to ride. If a freight elevator meets code requirements for passenger elevators, it may be certified to carry employee passengers. Each elevator so certified must have a sign stating this (see of *MS-1, Operation and Maintenance of Real Property* , Section 8 for additional information on certification). Freight elevators must have a sign with at least 2 inch tall lettering that indicates the maximum number of loaded mail containers allowed on the elevator.

2.1.2.3 Passenger Elevators

An elevator used primarily to carry persons other than the operator and persons necessary for loading and unloading.

2.1.2.4 Electric Traction Elevators

A power elevator where the energy is applied by means of an electric driving machine and which depends on traction between the hoist rope and drive sheave for hoisting the load.

2.1.2.5 Hydraulic Elevators

A power elevator where the energy is applied by means of a liquid under pressure, in a cylinder equipped with a plunger or piston. This is the most common type of hydraulic elevator found in postal facilities. It is a direction plunger (since the plunger or piston is connected directly to the car frame) electro-hydraulic elevator (since the pump that provided the pressurized liquid is driven by an electric motor). See also SECTION 8 – Hydraulic Elevators.

2.1.2.6 Sidewalk Elevators

An elevator of the freight type for carrying material exclusive of automobiles and operating between a loading zone in a sidewalk or other exterior area to a building and floors below the sidewalk or grade level.

2.1.3 Major Elevator Components

2.1.3.1 Electric Traction Elevator Components

2.1.3.1.1 Governor

The function of this device is to prevent over speed of the car. If the car speed exceeds a predetermined limit, the governor will first trip the overspeed switch and then set the safeties.

2.1.3.1.2 Safeties

The three types (A, B, & C) and various configurations of safeties which are described in Section 6.4 These devices grip the side rails and stop the car when actuated by the governor. It should be noted that they only operate when the car is moving downward.

2.1.3.1.3 Hoisting Machines

2.1.3.1.3.1 General

Traction hoisting machines depend on the traction between the sheave and the ropes for operation.

Winding drum machines have one end of the rope fastened to the drum. There are many disadvantages in winding drum machines, and few, if any, are installed in postal facilities.

2.1.3.1.3.2 A/C Hoisting Machines (Traction)

These use alternating current motors and reduction gears. These are cheaper to install and usually have two speeds. They are usually found only in older installations or in small seldom used locations.

2.1.3.1.3.3 D/C Geared Traction

Geared traction machine uses a direct current motor with a reduction gear between the motor and traction sheave. This is usually a worm gear and DC power is usually furnished by a motor generator.

2.1.3.1.3.4 D/C Gearless Traction

The hoisting sheave is connected directly to the motor shaft. DC power is usually furnished by a motor-generator. This requires a larger hoist motor but it is a very durable hoisting machine which can be accurately controlled. Many gearless machines that have operated for 30 years can have many more years of dependable service if controls and other equipment are upgraded.

2.1.3.1.4 Buffers

Buffers are in the pit to retard the downward motion of the car if it passes the bottom limit switch without stopping. The spring type buffer is used for car speeds less than 200 feet per minute only. Buffers are required under both the car and counterweight.

2.1.3.1.5 Roller Guides and Guide Shoes

Roller guides or guide shoes attach to the top and bottom of the car and guide the car along the guide rails. Roller guides are used for high speed applications.

2.1.3.1.6 DC Brake

DC brakes usually clamps on an extension of the hoisting sheave or driving shaft to prevent movement when loading or unloading. It should be noted that when DC power is applied, the brake is released and when the circuit is broken, the brake is applied.

2.1.3.1.7 Counterweight

A counterweight is usually made up of cast iron sections. One end of the hoisting ropes set is attached to the counterweight. To provide adequate traction between the ropes and driving sheave, the counterweight should usually equal the weight of the empty car plus 40% of the rated load. Too small a counterweight and poor traction will result in excessive wear of the ropes and sheave.

2.1.3.1.8 Motor Generator Set

Elevator motors using direct current electricity generally have an MG (motor generator) set to convert the AC to DC. Car speed is controlled by an individual generator for each elevator where the voltage applied to the driving machine motor is adjusted by varying the strength and direction of the generator field.

2.2 TYPES OF OPERATION

2.2.1 Automatic Elevator Operation

2.2.1.1 General

In general, automatic operation covers starting of the car in response to:

- Pushing operating devices at the landing.
- Operating devices in the car identified with the landings.
- An automatic starting mechanism where the car is stopped automatically at the landings.

2.2.1.2 Collective Automatic Operation

2.2.1.2.1 Nonselective

This is automatic operation by means of one button in the car for each landing level served and one button at each landing. All stops registered by pushing landing or car buttons are made, regardless of number or sequence of buttons pushed. The car stops at all landings where buttons have been pushed, in the order in which the landings are reached after the buttons have been pushed, but regardless of its direction of travel.

2.2.1.2.2 Selective

This is automatic operation by means of one button in the car for each landing level served and by up-and-down buttons at the landings. All stops registered by the pushing of the buttons are made as stated in Paragraph 2.2.1.2.1, but the stops registered by the pushing of the landing buttons are made in the order in which the landings are reached in each direction of travel. With this type of operation, all up landing calls are answered when the car is going up and all down landing calls are answered when the car is coming down, except in the case of the uppermost or lowermost calls which are answered as soon as they are reached.

2.2.1.2.3 Single Automatic Operation

This is automatic operation by means of one button in the car for each landing level served and one button at each landing, so arranged that if any car or landing button has been pushed, the pushing of any other car or landing operating button will have no effect until the response to the first button has been completed.

2.2.2 Car-Switch Operation

In this operation the movement and direction of travel of the car are directly and solely under the control of an operator using a manually-operated car switch or continuous pressure buttons in the car.

2.2.3 Continuous-Pressure Operation

This is operation using buttons or switches in the car and at the landings, any one of which may control the movements of the car as long as the button or switch is manually maintained in the actuating position.

2.2.4 Pre-Register Operation

This is operation where signals to stop are registered in advance by buttons in the car and at the landings. At the proper point in the car travel, the car operator is notified by a signal (visual, audible, or otherwise) to begin the stop, after which the landing stop is automatic.

2.2.5 Signal Operation

This is operation using single buttons or switches (or both) in the car, and up-or-down direction buttons (or both) at the landings. Known landing stops may be set up or registered for one or a group of elevators. The stops set up by pushing the car buttons are made automatically in succession as the car reaches those landings, regardless of its direction of travel or the sequence of buttons pushed. The stops set up by pushing the up and down buttons at the landing are made automatically by the first available car in the group approaching the landing in the responding direction, regardless of the sequence in which the buttons are pushed. With this type of operation, the car can be started only by means of a starting switch or button in the car.

2.2.6 Escalator Operation

Escalators are usually operated continuously while the building is occupied. In buildings where escalators are not installed and operated in pairs, they may be set to run in the direction of the flow of traffic (up, when the building is filling up; down, when the building is emptying). A schedule showing time for starting, reversing direction, and stopping of each escalator must be prepared and given to the shop which is assigned that responsibility. The Buildings Manager should retain a copy of the schedule. Guards, engineers, mechanics, etc., may be assigned to start or stop escalators, particularly if the operating schedule extends to the time when the person who would normally work on vertical transportation equipment is off duty.

2.3 SIGNAL SYSTEM AND EQUIPMENT

2.3.1 General

These consist of buttons or other devices located at the landings, which when pushed by a waiting passenger illuminate a flash signal or operate an annunciator in the car indicating floors where stops are to be made.

2.3.2 Types of Devices

2.3.2.1 Elevator Car Flash

This provides a signal light in the car that is lighted when the car approaches the landings where a landing signal registered device has been actuated.

2.3.2.2 Signal Registering Device (Elevator Landing)

This is a button or other device located at the elevator landings, which when actuated by a waiting passenger causes a stop signal to be registered in the car.

2.3.2.3 Signal Transfer Device (Elevator Automatic)

A signal registered in a car is automatically transferred to the following car if the first car passes without stopping at floor where a signal has been registered.

2.3.2.4 Signal Transfer Switch (Elevator)

This is a manually operated switch located in the car, by which operators can transfer a signal to the next car approaching in the same direction if they wish to pass a floor where a signal has been registered in the car.

2.3.2.5 Starter's Control Panel (Elevator)

This is an assembly of devices where the starter may control the way in which elevators function.

2.3.2.6 Terminal Stopping Device (Normal)

This is a device to slow down and stop an elevator automatically at or near a terminal landing independently of the functioning of the operating device.

2.3.2.7 Terminal Stopping Device (Final)

This is a device which automatically causes the power to be removed from an electric elevator driving machine motor and brake, independent of the functioning of the normal-terminal stopping device, after the car has passed a terminal landing.

2.3.2.8 Traveling Cable

This consists of electric conductors which provide electrical connection between an elevator or dumbwaiter car and fixed outlet in the hoistway.

2.3.2.9 Selector Tape

This is a metal tape between the car and selector, which drives the selector according to the position of the car.

2.4 MEASURING OPERATING PARAMETERS

2.4.1 Application

The following measures of value should be applied to every new or modernized installation:

- Time to destination.
- Average interval of departure.
- Handling capacity.

2.4.2 Measures

2.4.2.1 Time to Destination

The first measure, time to destination, is a summation of average waiting time and average traveling time, obtained from a series of traffic studies in representative buildings. For an office building, the time to destination should not exceed 45 seconds.

2.4.2.2 Average Interval of Departure

The second measure, average interval of departure, is the average time between the departure of elevators from the main lobby floor. A suitable interval of departure for a large city office building is generally considered to be between 20 and 30 seconds. This interval of departure, with other factors as to speed and capacity properly selected, will enable rapid handling of passengers during the peak periods of traffic.

2.4.2.3 Handling Capacity

The third measure, handling capacity, involves the individual capacity of each elevator, the total number of people in the building, and the number of elevators to be provided.

2.5 SIGNAL CONTROLS FOR AUTOMATIC ELEVATORS

2.5.1 Basic Control or Coordinating Systems

2.5.1.1 Collective Control

2.5.1.1.1 General

Collective control is designed for not more than three elevators. It is directly responsive to traffic demand; its application is in apartment buildings and small, lightly populated buildings where heavy concentrated peak traffic periods are rare. The advantage of this type of system is that it is simple and economical to install.

2.5.1.1.2 Characteristics

The collective control has these inherent characteristics:

- Elevators have a tendency to travel in close proximity to one another.
- Rapid elevator shutdown after calls are answered.
- Perpetual highest call reversal.

All three of these characteristics result in excessively long waits for an elevator. Hence, the system is not acceptable for high-rise multi-occupancy office buildings but it is satisfactory for small buildings and often for freight elevators.

2.5.1.2 Automatic Operator Less Control

2.5.1.2.1 General

The second basic elevator coordinating system is known generally as the automatic operator less control and is designed for a group of three or more elevators. It employs time dispatching intervals between cars.

2.5.1.2.2 Subdivisions

There are many subdivisions to this basic system plan. Two main ones are:

- A system designed to maintain a schedule, with cars dispatched from both top and bottom terminals at timed intervals.
- A system which does not attempt to maintain a schedule, but dispatches cars at random, according to service demands.

2.5.1.2.3 Random Dispatching

The term random dispatching, as used here, applies to those systems which dispatch an elevator away from the terminal floor after a non-variable, pre-determined waiting time. If more than one elevator were at the dispatching terminal, this time for the second and subsequent cars would be measured from the time of departure of the previous car. Usually this is a 10-15 second interval.

2.5.1.2.4 Scheduled Dispatching

An elevator dispatching system with regularly scheduled dispatching intervals attempts to dispatch the cars away from the terminal floors at time intervals. It automatically adjusts the time intervals to coordinate the movement of all elevators to maintain a balanced flow. In contrast to the random dispatching system, an elevator may wait at the dispatching terminal 45 seconds if the elevator arrived at the terminal early or if the next car is late in arriving. On the other hand, if the first car was late arriving at the terminal and the second car was early, the waiting time of the first car may be reduced to 10 seconds.

In addition, the dispatching signals at the top and bottom terminals in some systems are coordinated so that if an elevator is late in leaving the bottom terminal, the signal at the top terminal will be delayed. Consequently, the elevators are equally spaced in time when they leave the terminals.

In order to achieve a more equal spacing of elevators, however, one elevator will be intentionally delayed from leaving the top or bottom terminal if it has completed the prior trip making fewer than the average number of stops. Therefore, the average interval between elevators may be several seconds longer than that obtained with a random dispatching system.

SECTION 3

CLEANING AND LUBRICATING

3.1 CLEANING ELEVATOR EQUIPMENT

3.1.1 Reasons for Cleaning

Cleaning of both mechanical and electrical components is an essential part of the elevator preventive maintenance program, because this:

- Helps disclose damaged or worn parts. Dirt and grease build-up will hide damaged or worn parts until the failure creates a danger to life or expensive repairs.
- Prevents fire hazards. Dirt will collect moisture and create short circuits; oil and grease will ignite easily. The build-up of oily lint on elevator rails and trash in pit is an extreme fire hazard. If a fire occurs in the elevator hoistway, persons in the car may be killed by the heat or smoke.
- Helps prevent safety hazards. Oil and grease cause slipping hazard; loose tools or spare parts create hazards.
- Reduces equipment failures. Dirt in moving parts causes excess wear. Dirt, oil, or grease between electrical contacts causes excess arcing and increases electrical resistance of contacts. Dirt and grease buildup also act as insulation and cause overheating. Because of this and for safety considerations, elevator equipment must be kept clean.

3.1.2 Cleaning Materials

3.1.2.1 Solvents

If necessary to use a solvent to remove oil or grease, use only approved safety types and provide adequate ventilation. The following solvents listed in the GSA Catalog may be used:

- Liquid solvent cleaner for cold cleaning operations-NSN 6850-00-619-7715.
- Safety solvent (standard compound)-FSN 6890-00-597-9765

The following solvents are highly toxic and should not be used for cleaning:

- Carbon tetrachloride
- Benzol
- Gasoline
- Naptha

Other solvents purchased locally should only be used for special cleaning problems. In each case, follow the label on the container explicitly. The importance of adequate ventilation when using any solvent needs to be continually stressed. The use of solvent in hot weather or to clean warm machinery increases the evaporation rate and ventilation requirements. Avoid continuous or repeated contact of solvents with the skin. Use neoprene coated gloves or aprons to provide protection against skin contact.

3.1.2.2 Wiping Cloths

Lint free cloths or disposable wipes should be used. Waste or cloths which leave lint should not be used, as the lint will adhere to the insulation and increase the collection of dust, moisture and oil.

3.1.3 Methods

3.1.3.1 General

Before cleaning elevators, the main switch should be open, tagged, and locked. When the cleaning job is complete, check the operation of the elevator to assure that the cleaning has not inhibited the movement of a mechanical part or caused a relay or contact to become inoperative. The proper cleaning of relays and contacts is covered in SECTION 5.

3.1.3.2 Mechanical Parts

Clean mechanical parts by wiping with a lint free rag. If there is a heavy build-up of oil or grease, an approved solvent should be used. In some cases, a strong vacuum cleaner should be used prior to wiping. This is especially true of hoistway walls, guide rails, safeties, etc. where there may be a rapid build-up of paper dust from mail handling operations. It is very important that this not be allowed to accumulate because of the extreme fire hazard of paper dust soaked with oil.

Note that when dust accumulates on ropes, it tends to absorb the lubricant and complicate the problem of providing the proper amount of rope lubricant.

When guide rails require lubrication, they must be thoroughly cleaned, removing old lubricant at least once each year.

3.1.3.3 Electrical Components

The preferred method of cleaning electrical components is with a strong vacuum, and brush attached to a non-conductive wand. This may be used on:

- Motor winding,
- Around brushes and brush holders,
- Relays, and
- Control boards.

In some instances, compressed air (below 30 psig) may be needed. However, the use of compressed air in most instances is discouraged since it simply redistributes the dirt and some is bound to settle on the contacts. When compressed air is used:

- Always vacuum afterwards,
- Clean contacts with a soft dry lint free cloth; and
- Check the operation

The removal of the build-up of dirt and grease on windings and insulation, which requires solvents or blasting with mild abrasives such as ground nut shells, is the work of specialists and is not covered here.

3.2 LUBRICATION

3.2.1 General

Proper lubrication is a very important function of maintenance. Note that the key word in this statement is proper. The tendency to over-lubricate, lubricate too frequently, lubricate an easy way, or use any lubricant, is often the cause of unidentified machine failure. Therefore, this subchapter deals with proper lubrication of various elevator components.

3.2.2 Bearings

3.2.2.1 Antifriction Bearings

Antifriction ball and roller bearings fail most often by over or improper lubrication rather than inadequate lubrication. When these bearings are located in a gear case, oil from the gear case will usually provide sufficient lubrication. Bearings on motors or extended shafts usually have pressure type fittings.

3.2.2.2 Sleeve Bearings

Sleeve bearings lubrication is very critical since the absence of lubricant will result in metal-to-metal contact between the shaft and the bearing. This will cause the bearing to fail in only a few minutes. Therefore, where sleeve bearings are used, a means of supplying lubricant is always provided and care must be taken to assure that the reservoirs are always adequately filled with the correct lubricant.

3.2.3 Worms and Gears

3.2.3.1 Description

The most common type reduction gear used on elevator equipment is where the motor shaft is connected to the worm and the traction sheave to the gear. This provides a high reduction (the ratio of the number of teeth on the gear to the number of starts on the worm) and is very difficult to drive backwards. i.e., the gear to the worm. There are, however, rubbing surfaces under heavy pressure where the worm and gear contact.

3.2.3.2 Oil Characteristics

The oil used in the gear case must have sufficient body to prevent metal-to-metal contact on the gear teeth and also provide proper lubrication for the bearings. The correct level of the oil is also very important. Some manufacturers arrange their machines so that only the lower side of the worm dips in the oil. If the oil level is allowed to fall below the worm, metal-to-metal contact and rapid wear will occur between the worm and the gear. Also, if the case is overfilled, the worm will act as a pump, cause excessive foaming, increase energy consumption, overheat the oil, reducing its lubricating qualities, and usually cause messy leaks. Therefore, maintaining the correct oil level is very important.

3.2.4 Suspension and Compensating Wire Rope Lubrication

The ropes shall be kept lightly lubricated and clean in order to readily permit visual inspection. It is imperative that the ropes are not over-lubricated in order to avoid the loss of traction. Further, it is essential that all ropes in the set are equally tensioned.

3.2.4.1 Reasons for Lubricating

A wire rope is composed of a number of wires twisted to form a strand: several of these strands are then twisted around a hemp center to become a rope. As the rope bends around sheaves, the wires and strands rub against each other and therefore require lubrication. Load variations, acceleration, size of sheaves, etc. also cause rubbing action requiring lubrication. When the rope is manufactured, the hemp center is saturated with a lubricant. This provides sufficient lubrication for a considerable period (2 or 3 years) depending on the amount of operation, size of sheaves, load, atmosphere, etc. An inadequately lubricated rope will cause internal friction in the wires, increase stress, and result in broken internal wires. Since these broken wires cannot be observed by inspection, a dangerous condition is created. Likewise, over-lubrication is messy, causes poor traction, and excessive wear between the ropes and the traction sheave.

3.2.4.2 Governor Wire Ropes

Governor wire ropes shall be kept clean, however unlike suspension ropes, they shall NOT be lubricated because the lubricant may interfere with the ability of the governor jaws to hold the governor rope. In the event they have been lubricated, the lubricant shall be removed. If the lubricant cannot be removed, the governor wire rope shall be replaced and the governor and safety tested as per ASME A17.1.

3.2.4.3 When to Lubricate

A practical test for the need of lubrication is to wipe a finger in a sheave groove (with power off). This should leave a faint smudge and have a slightly oily feel. If this test leaves the finger dry and clean, lubrication is advisable.

3.2.4.4 What to Use

The lubricant and amount used shall be limited to that supplied or approved by established elevator or wire rope manufacturer.

3.2.4.5 How to Apply

The lubricant may be applied with a brush or with a lubricating device that applies the lubricant as the ropes travel through it. Prior to applying the lubricant, the ropes should be cleaned using a solvent if necessary to remove dirt and dust accumulation. Sufficient lubricant should be applied to penetrate to the inside strands. Take care to catch drips while applying the lubricant and clean up afterward.

3.2.4.6 Check after Lubrication

Since excessive or improper lubricants will seriously reduce available traction, the traction sheave should be carefully observed after lubrication to determine if there is excessive slippage during acceleration or retardation.

3.2.5 Guide Rail Lubrication

3.2.5.1 Proper Lubrication

Proper guide rail lubrication is most difficult. Too much or the wrong type of lubricant will cause improper operation of safeties. The lubricant also collects dust and creates a serious fire hazard. Therefore, lubrication shall be in compliance with the manufacturers specifications indicated on the crosshead data plate. The data plate is attached to the door crosshead in an easily visible location stating:

If the plate is not provided, lubrication shall be as follows:

- Guide rails shall be kept lubricated. The exception to this is those guide rails equipped with rollers or other types of guiding members.
- If a sliding-type safety is used, the guide rail lubricant, or the pre-lubricated or lubricant-impregnated guide-shoe gib, shall be in accordance with the safety device manufacturer's specifications.

If the manufacturer's specifications concerning lubricants are not followed, the safety device shall be tested in accordance with ASME A17.1 in order to demonstrate that it will function as intended.

3.2.5.2 Cleaning Guide Rails

Guide rails must be kept clean and free from excess lubrication as this will accumulate lint and dust. A method to collect excess lubricant shall be provided at the bottom of the rails.

CAUTION

Compounds designed to prevent rust (paint, mixtures of graphite and oil, etc.) unless specifically recommended by safety's manufacturer, shall not be used on the guiding surfaces. If they are applied, the safety must be checked in accordance with ASME A17.1.

Rails on which lubricant is used must be cleaned at least once per year. A solvent is usually necessary to remove excess lubricant, lint, dust, and paper dust which create a serious hoistway fire hazard.

3.2.5.3 Rubber Guide

The rubber tire roller guide is used in many new installations. These have many advantages, including the elimination of the need for rail lubrication and reduction in elevator operation power requirements.

3.2.6 Oil Buffers

The oil level in the buffers should be checked at least once per month. The grade of oil and level of the oil in the buffer shall be maintained as specified by the manufacturer's indicator and specification. Consult the Buffer Marking Plate for the:

a. Maximum and minimum loads and maximum striking speeds
b. The permissible range in viscosity of the buffer oil to be used
c. The viscosity index number
d. The pour point in degrees Celsius (Fahrenheit)
e. The stoke of the buffer
f. The composition of the gas
g. The name, trademark or file number of which the organization that manufactured the product can be identified
h. The certification markings

Buffer plungers shall be kept clean and shall not have anything applied to them that might interfere with their intended operation.

3.2.7 Controller Contactors and Relays

Lubricate these items only where necessary, as specified by the manufacturer. These items must be kept clean and free from dirt. Therefore, carefully avoid over lubrication.

SECTION 4

ELEVATOR DOCUMENTATION

4.1 GENERAL

Site managers must preserve and maintain copies of all elevator electrical schematics, mechanical drawings, parts lists, design specifications, inspection documentation, and any other pertinent information. Refer to current Maintenance Instruction or MLB for vendor requirements in scenarios where documentation is not available.

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SECTION 5

MECHANICAL/ELECTRICAL DEVICES

5.1 CONTACTS AND RELAYS

5.1.1 General

5.1.1.1 Background

Relays and contacts are often found in elevator control systems. Some redundancy is designed in the control circuits for safety in the event of component failure. Basically the system is designed to ensure the failure of any single magnetically operated switch, contactor, or relay to release in the intended manner, or the occurrence of a single accidental ground, shall not permit the car to start or run if any hoistway door interlock is unlocked or if any hoistway door or car door or gate contact is not in the closed position.

5.1.1.2 Restriction

Some maintenance technicians have found that some relays can be jumped with a wire and an elevator will operate (that is, until another relay fails and causes the car to run into the pit or overhead, or run with the doors open because the relay that was jumped could not perform the safety function designed for it in the circuit). Elevators must never be operated with relays jumpered in this manner.

5.1.1.3 Purpose of Maintenance

The more relays in a circuit, the higher the probability of a failure. However, many of these failures can be eliminated by good maintenance. This subchapter will deal with general requirements for relay and contact inspection and maintenance. This is not intended to replace manufacturers' instructions but to provide one with the knowledge to better understand and interpret those instructions.

5.1.2 Contacts

5.1.2.1 Contact Resistance

Since a contact is an electrical conductor, one of the most important characteristics is contact resistance. The contact resistance can be expressed by the following formula:

R=K/P

Where:

R is contact resistance,

K is factor for the contacts depending on the design of the contacts and the conductivity of the material from which the contacts are made,

P is the contact pressure (force holding the contacts together).

From this, it is obvious that the larger the contact pressure, the smaller the contact resistance. The contact pressure is part of the relay design and should not be increased, but it is important that it not be reduced from its design value. Checking for correct spring pressure should be included as part of relay maintenance. Contact pressure may be reduced because of worn contacts or weak springs. Contact pressure should be checked and adjusted to the manufacturer's specification.

5.1.2.2 Contact Material

5.1.2.2.1 General

A great number of contact materials are in use in the electrical industry. However, the three most widely used materials in the elevator industry are graphite (or carbon), silver, and copper.

5.1.2.2.2 Carbon

Carbon contacts have been used for many years, especially where large currents are switched, because carbon will not weld. Therefore, carbon contacts are less likely to stick closed and consequently safety factors of the relay are improved. Carbon contacts also exhibit a higher resistance to arcing.

Carbon contacts do wear and are usually made with changeable inserts. Care must be taken that the carbon inserts are not allowed to wear away resulting in metal to metal contact and almost certain sticking of the relay due to welding.

5.1.2.2.3 Copper

5.1.2.2.3.1 Characteristics

Copper is a very good conductor and has been used in electrical equipment for many years. However, when copper is exposed to air, copper oxide will form and copper oxide is a poor conductor. The process of forming copper oxide is very slow at normal room temperature but as the temperature increases, the formation of copper oxide is accelerated. Therefore, overheating will increase copper oxide formation and contact resistance. This results in more overheating and resistance until failure occurs.

5.1.2.2.3.2 Requirements

Copper contacts in power circuits should be examined regularly as part of the preventive maintenance program. Contacts that operate frequently are not as likely to develop a copper oxide coating as those that remain open or closed for large periods. The mechanical rubbing action of operation usually breaks up the formation of oxide. Both power and control circuit switches, which remain closed for long periods of time, should be operated periodically to break up the oxide formation.

5.1.2.2.3.3 Silver

Silver and silver alloys are excellent conductors. Silver oxide is also a good conductor. Therefore, silver contacts on elevator control circuits should require no maintenance unless they are contaminated with foreign matter such as oil or dirt. Silver contacts are often used on very low voltage and a small amount of dirt or oil will prevent proper operation. If silver contacts require cleaning take, special care to assure that the silver is not removed.

5.1.2.3 Contact Shape

Switch and circuit breaker contacts are of various shapes and usually have a rubbing action when making or breaking. However, relay contacts are usually flat or convex as shown in Figure 5-1. In some cases, both contacts are flat while others have one flat and one convex contact. The original shape of the contact must be retained during cleaning. If burning or pitting has distorted the contact so that it needs reshaping, the original shape should be restored. The person responsible for relay maintenance should become familiar with the proper shape of the contacts. Note that some silver contacts are solid while others are silver-plated. Any attempt to reshape silver plated contacts is useless. When contacts are replaced, both the moving and stationary contacts should be replaced.

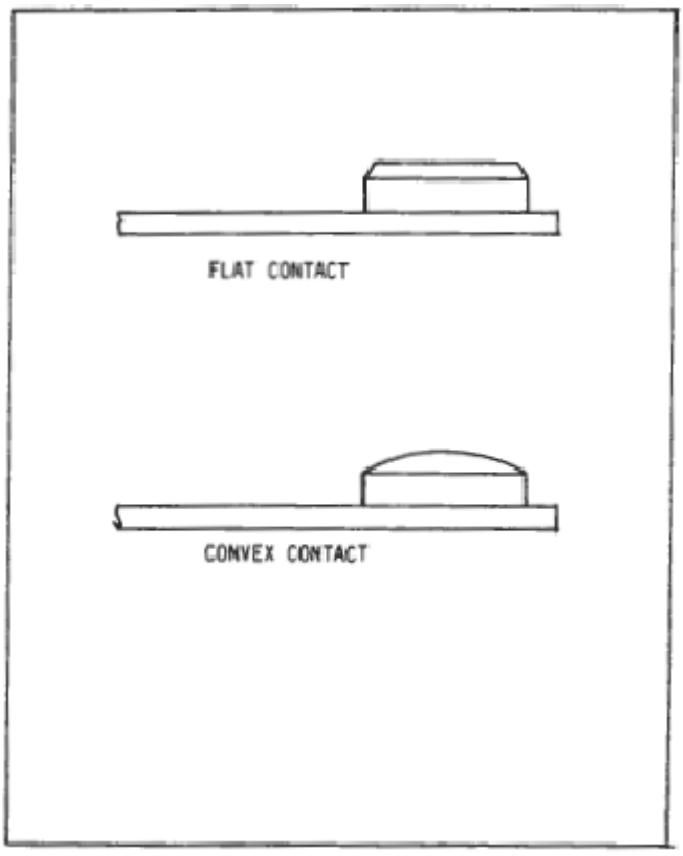


Figure 5-1. Relay Contact Shapes

5.1.2.4 Arc Rupturing Parts

When contacts are carrying currents that are difficult to interrupt, they are equipped with arc quenching devices. These devices usually surround the contacts in a very definite position. These devices are usually removable for inspection and cleaning and it is very important that they be replaced in the correct position. The devices or shields on DC equipment are usually molded and should be replaced before the molded material is burned away exposing metal parts. The AC contacts use the de-ion arc quencher as shown in Figure 5-2. As the specially shaped contacts separate, a magnetic reaction causes the arc to rise into the de-ion grid. At the next zero point on the current cycle the air is de-ionized and the rising voltage is insufficient to re-ionize the air and the arc is quenched. Pitting and burning of the contacts is greatly reduced with this de-ion system.

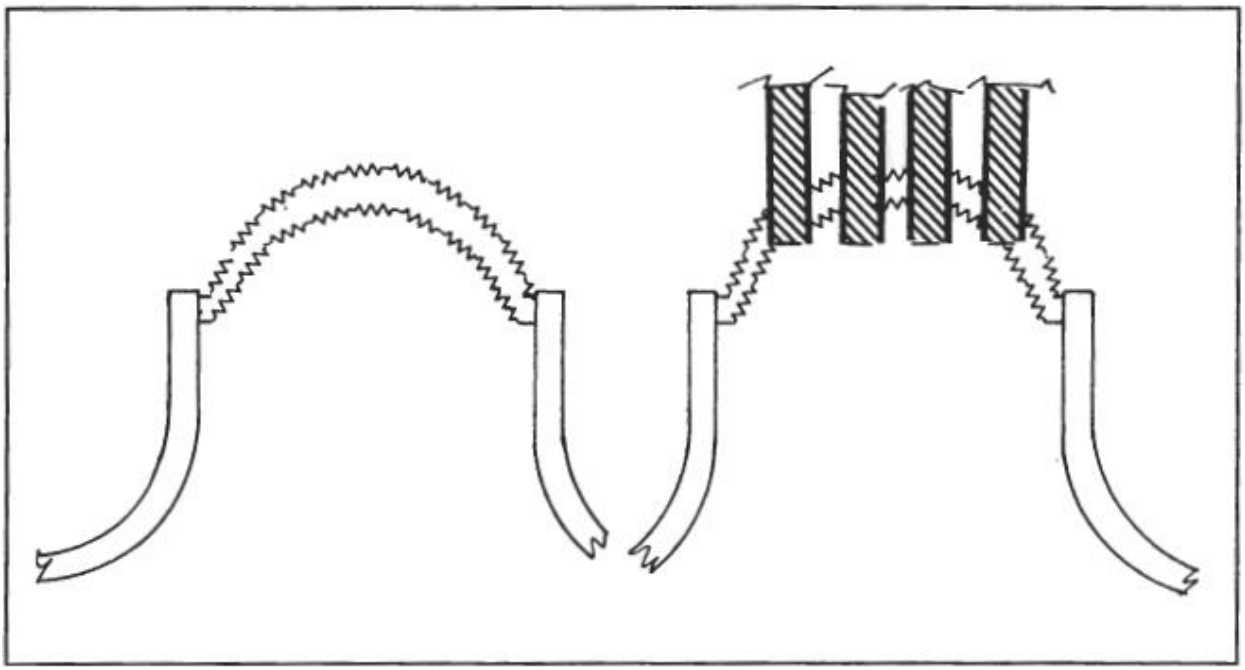


Figure 5-2. AC Contact with De-ion Arc Quencher

5.1.2.5 Maintenance of Relay Contacts

5.1.2.5.1 Hard Alloy Contacts

Hard alloy contacts may be cleaned by drawing a clean lint-free cloth or smooth paper between them while they are manually closed. If it is necessary to moisten the cloth with cleaning fluid, use a dry cloth or paper strip for polishing. A burnishing tool may be used to clean corroded, burned, or pitted contacts. A contact should not be burnished, however, unless it is found to be pitted or oxidized and then not any more than is necessary to restore a smooth, clean surface. The original shape of the contact must be retained.

5.1.2.5.2 Silver Contacts

Silver contacts may be cleaned with a cloth or brush moistened with contact cleaning fluid. After cleaning, polish contact with a dry lint-free cloth. The brown discoloration on silver contacts is silver oxide, which is a good conductor and should not be removed. If solid silver contacts are deeply pitted, they may be reshaped with a fine file or #0000 sandpaper. The sandpaper may be drawn through the contacts while they are gently pressed together. If the shape is altered, it should be reshaped. After this, the contact must be thoroughly cleaned with soft cloth to remove all foreign matter such as sand. Silver plated contacts may be cleaned with a soft lint-free cloth moistened with cleaning fluid and then polished with a soft lint-free cloth. This must be done very carefully to avoid removal of too much plating. If silver plated contacts are pitted or burned, they should be replaced. After cleaning, the position of contact arms and contact pressure should be checked and adjusted to manufacturer's specifications.

5.1.3 Relays

5.1.3.1 Description

Many types and sizes of relays are used in elevator circuits. A relay is a device that is controlled from an external source and in turn controls some other device. This device may be the motor generator, door operator, safety switch, etc. In some instances, two or more relays are in series and all must close before the controlled device can operate. A relay in which the contacts are open when its coil is not energized is referred to as normally open and a relay in which the contacts are closed when its coil is not energized is referred to as normally closed. There are also relays to perform special functions such as thermo-overload relay, reverse phase relay, time delay relays.

5.1.3.2 DC Relay

Relays can be designed to be operated by either AC or DC power. The designs of the AC and DC relays are quite different and they are not interchangeable. Figure 5-3 shows an outline of a DC relay. When current flows through the coil, a magnetic field is generated and pulls the armature (sometimes called a clapper) against the pole piece. The strength of the magnetic field is proportional to the number of turns in the coil and current flowing in the coil. The dash lines in Figure 5-3 show the path of magnetic flux. The amount of current flowing through the coil is determined by the resistance of the coil and is independent of the position of the armature. When the coil is de-energized, residual magnetism will tend to hold the armature against the pole piece causing the relay to stick. Therefore, a thin piece of nonmagnetic material is usually placed on the end of the pole piece to hold the armature away and prevent this sticking. A thickness of .002 or .003 inch is usually sufficient to prevent sticking.

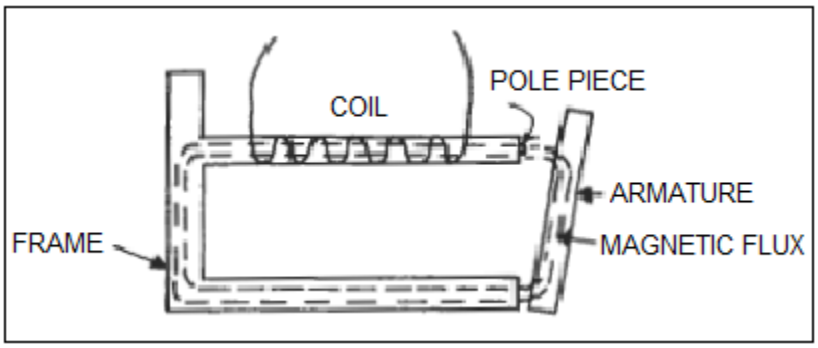


Figure 5-3. DC Relay

5.1.3.3 AC Relay

As the name indicates, the AC relay is operated by alternating current, usually single phase. The design and construction of the AC relay is quite different from the DC relay. The frame and armature are laminated to reduce eddy current losses. Since AC voltage and current pass through a zero point 120 times each second (60Hz), and the laminated frame loses residual magnetism very rapidly, the armature will be jerked 120 times each second causing a buzzing sound. This noise is eliminated by a device called a shading coil. This is simply a heavy conductor (usually copper) in the shape of a "D." The straight part of the D is placed in a slot in the pole piece.

The remainder of the shading coil extends to the outside of the pole piece as shown in Figure 5-4. This has the effect of inducing a secondary magnetic field and eliminates the buzzing noise. If this pole shading coil breaks, the relay will buzz.

The coil of the AC relay is also quite different from the coil of the DC relay. The DC coil consists of many turns of fine wire and the resistance of the wire regardless of the armature position limits the current flow. The AC coil consists of relatively few turns of large wire and the current flow is limited by impedance. When the magnetic circuit is open, the impedance is very small and when the magnetic circuit is closed (the armature tight against the frame) the impedance is high. Therefore, if the armature is blocked or for any other reason (such as under voltage) does not close against the frame with zero gap the current will be excessive, overheat the coil and cause failure.

Therefore, the pick-up or drop out of AC relays cannot be time delayed by devices that restrain the action of the armature. Battered laminations due to repeated striking of the armature will also cause poor alignment and premature coil failure.

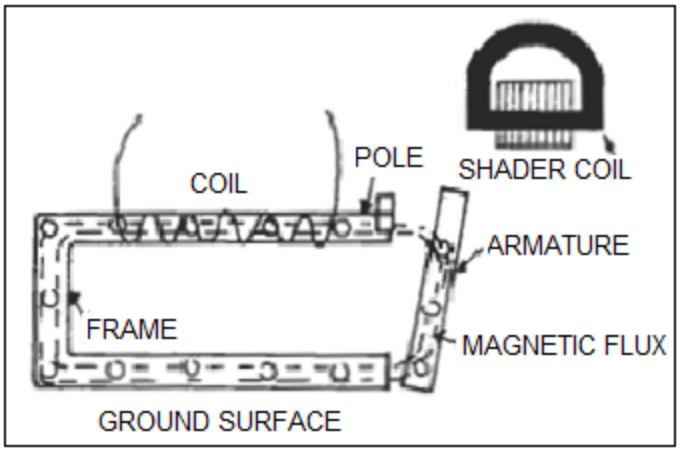


Figure 5-4. AC Relay

5.2 BRUSHES AND COMMUTATORS

5.2.1 General

Care of brushes and commutators is perhaps more critical than any other area of elevator maintenance. Clean commutators and free operating brushes are essential for proper leveling and good commutation. One of the greatest sources of commutation trouble is oil from bearings forming a film on the commutators. This also attracts dirt, causing the brushes to stick and/or short circuit. The brushes and commutators are simply moving contacts and proper care is necessary to minimize electrical resistance without causing excessive wear.

5.2.2 Commutators

5.2.2.1 Construction

The commutator is made up of copper bars held together with a "V" ring and usually insulated from each other and the frame with mica.

5.2.2.2 Problems with Commutators

5.2.2.2.1 High Bars

If one or two commutator bars (or segments) are higher than others, the brushes will jump and spark. The bars may be high at either end or across their entire lengths. This can also be caused by loose bars. Either of these conditions will cause a progressively longer spark and damage adjacent bars. High or loose bars can be identified by listening for brush noise or placing a wooden stick on the brush to feel the vibration. If the condition is caused by a loose bar, the clamping ring should be tightened. Otherwise, the high bars should be corrected by grinding or turning the commutator.

5.2.2.2.2 Flat Spots

Flat spots prevent proper contact of the brushes causing sparks and damaging the commutators. These may be caused by-

- Improper undercutting.
- High mica.
- High bars.
- Low bars.
- Burnt bars (due to poor commutation).

This can be corrected by undercutting the mica and/or grinding or turning the commutator.

5.2.2.2.3 High Mica

The insulating mica between each bar should be undercut so that it does not interfere with the brushes contacting the commutator. If the mica extends above the commutator surface, the brushes will be noisy and excessive arcing will occur. Always consult the manufacturers' instructions for proper amount and shape of undercut.

5.2.2.2.4 Eccentricity

An out-of-round commutator will move the brushes in the holder with each revolution. This will cause excess side wear of the brushes in the holder. If the equipment is turning at high speed, the brush contact will also be poor and cause arcing during part of the revolution. This can also be caused by bad bearings. The only solution is to replace the bearing and/or turn the commutator.

5.2.2.2.5 Grooving

This is simply wear of the commutator by the brushes being in one position. This leaves a groove between each set of brushes and is caused by improper staggering of brushes, and/or brushes that are too abrasive. It will result in poor commutation and excess arcing, especially when new brushes are installed. The remedy is to turn the commutator and replace or adjust the brushes to remove the cause.

5.2.2.2.6 Threading

This is small grooves cut randomly across the commutator. This may be caused by poor brush quality, inadequate brush pressure, or a contaminated atmosphere. Some industrial solvents are damaging to the commutator film. Fresh paint near the equipment may also cause this problem.

5.2.2.3 Care of Commutators

5.2.2.3.1 Surface Film

The commutator should not be a shiny copper but should be a smooth glassy brown, the color being somewhere between straw and milk chocolate brown. This indicates a surface film which is essential for good operation. This film is often 1,000 times thinner than a human hair and is made up of copper oxide, graphite, and water vapor. The film is caused by the rubbing action of the brushes and current flow between the brushes and commutator. The film can form in a few hours or may take several days.

If the film does not form, check for the following causes:

1. Too low current density. Each brush has a current density range in which it works best. If the current density is too low, adequate film will not form. Also, if it is too high, excess film may form and cause arcing and bar burning. Electrographic brushes usually operate best at 50 to 80 amps per square inch. Where multiple brushes are used, it may be necessary to remove a brush from each holder to obtain sufficient current density to establish a film. If this is done, remove a brush in the same position on each holder.
2. Insufficient brush pressure. Adjust to correct pressure.
3. Low humidity. This may occur in extremely dry climates and can be corrected by placing a container of water or humidifier near the equipment.
4. The presence of paint vapors or other fumes. Many vapors will attack and destroy the commutator film. Care should be taken to prevent fumes from entering the machine room. Some of the more common substances that harm the film are fluorine, chlorine, bromine, iodine, ketones, alcohols, esters, turpentines, paint fumes, tobacco smoke, and various cleaning compounds.

Care of the film on the hoist motor is very important since the leveling voltage is usually low and poor commutation will result in erratic leveling. Oil or other contaminates may be cleaned from the commutator using a locally procured cleaning device.

5.2.2.3.2 Clamping Ring

Loose clamping rings may cause sparking and loose or high bars. These bolts should be checked when the commutator is hot and tightened diametrically (i.e., after tightening one, proceed to the one on the opposite side of the shaft). Follow the manufacturer's torque specifications and use a firm steady pull. Excessive or careless tightening can damage the insulation or cause other failures.

5.2.2.3.3 Turning and Grinding

Turning a commutator is quicker than grinding and is required if the commutator is out of round or contains badly burned segments. However, turning requires removing the armature and setting it up in a lathe. This is beyond the scope of most of our in-house maintenance and should normally be contracted out. Grinding can be accomplished without disassembly, so it is used most often to correct commutator troubles. However, grinding is usually not considered part of day-to-day maintenance and it may be advantageous to contract it out also. The commutator may be ground with a hand stone or a special grinding tool. The special grinding tool is recommended because it is easier to use and can correct minor eccentricity which is only aggravated by a hand stone. Before turning a commutator, check its eccentricity with a dial indicator following the manufacturers' specifications. For generators the maximum allowable eccentricity is usually .002 inch and no more than .0001 inch from segment to adjacent segment.

5.2.2.3.4 Undercutting

This term refers to the removal of some of the insulating material (mica) between the commutator segments. Most medium and high-speed machines require that the mica between the segments be undercut. Undercutting on low speed motors is not always desirable, since the centrifugal force may not be great enough to dislodge foreign matter from the slots. Some new motors may also use a new type of insulation which does not require undercutting. If the commutator is furnished undercut by the factory, it should normally be undercut when repairs are made. Figure 5-5 shows the standard "U" and "V" slot shape used in undercutting. "U" slots are usually used in high-speed machines which operate in clean environments. "V" slots are usually better in slow speed machines and/or dirty environments since dirt and foreign matter is more easily dislodged from the "V" slots by centrifugal force.

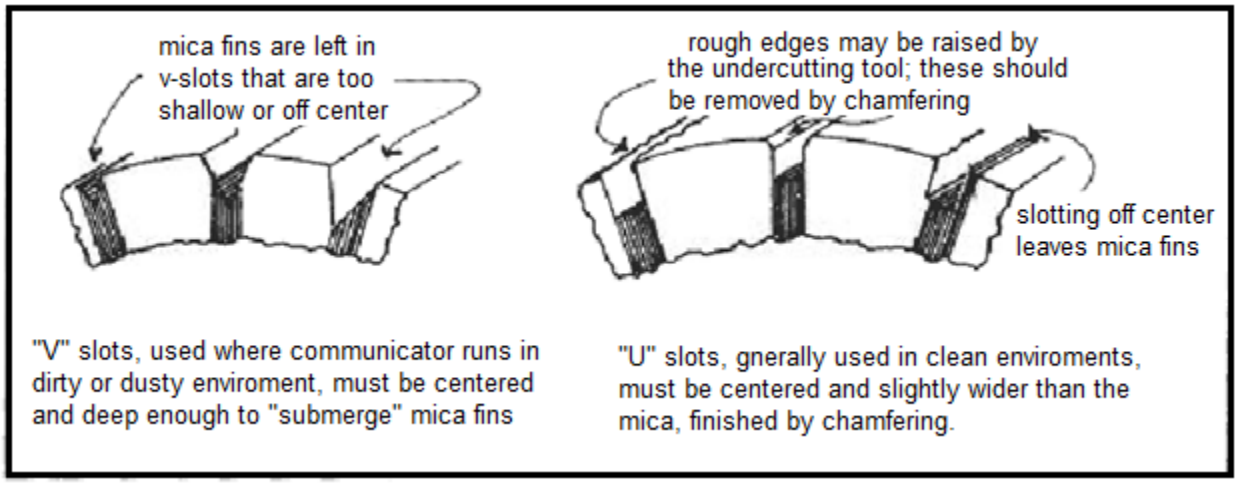


Figure 5-5. Commutator Undercut

5.2.3 Brushes

5.2.3.1 Brush Types

5.2.3.1.1 General

There are four basic types of brushes made for particular applications. The types and grade of brush specified by the manufacturer should be used unless a problem develops. The discussion of brush types will help evaluate brush problems.

5.2.3.1.2 Carbon Brushes

These brushes are inexpensive and are generally used for low speed low current density applications. These brushes give good performance where it is impractical to undercut the commutator, where a cleaning action is needed at low speeds and/or where strong oxidizing fumes are present.

5.2.3.1.3 Graphite Brushes

These are made from a mined material as contrast with carbon brushes which are manufactured from base materials. Graphite brushes can be identified by a flaky structure and silvery appearance. High temperatures are not used in manufacturing these brushes. Consequently, any desired cleaning material can be added to the graphite.

5.2.3.1.4 Graphite Carbon or Carbon Graphite Brushes

As indicated, this is a mixture of the two materials with the first material being the dominant ingredient. The materials are powdered, pressed together, and baked at low temperatures.

5.2.3.1.5 Electrographitic Brushes

The carbon brush is baked at high temperature in an electric furnace. This graphitizes some of the material and changes the characteristics. This type of brush has low friction; is non-abrasive, cool running, can carry more current than carbon, and is resilient (resist bounce). Since it is nonabrasive, it has little or no cleaning action.

5.2.3.1.6 Metal Graphite

The metal is usually copper. The materials are powdered, pressed together, and baked. The brushes are usually used on slip rings and commutators having high current and low voltage. These are normally not found on elevators.

5.2.3.2 Seating and Alignment

5.2.3.2.1 General

Proper seating and alignment of the brushes is very critical. The manufacturers' instructions and adjustment specifications should be carefully followed. See Paragraphs 5.2.3.2.2 thru 5.2.3.2.5 for aid in understanding the required adjustments.

5.2.3.2.2 Angle of Brush

The angles of the brush mounting in relation to commutator radius are shown in Figure 5-8. The radial mount is usually used on reverse operation such as hoist motors. However, at least one elevator manufacturer has developed a brush holder that allows the leading or trailing mount to be used on the hoist motor. This mount tends to hold the brush tight against one side of the brush holder. Note that the trailing mount has a maximum angle of 20 degrees while the leading mount has a minimum angle of 30 degrees.

5.2.3.2.3 Clearance

The clearance between the brush holder and the commutator is shown in Figure 5-7. Too great a clearance will cause brush chatter and sometimes break the brush. The clearance between the brush and the brush is normally held between .002 and .001 inch. Correct clearance is very critical on reverse rotation, since this will rack a loose brush back and forth. This reduces face contact and causes arcing.

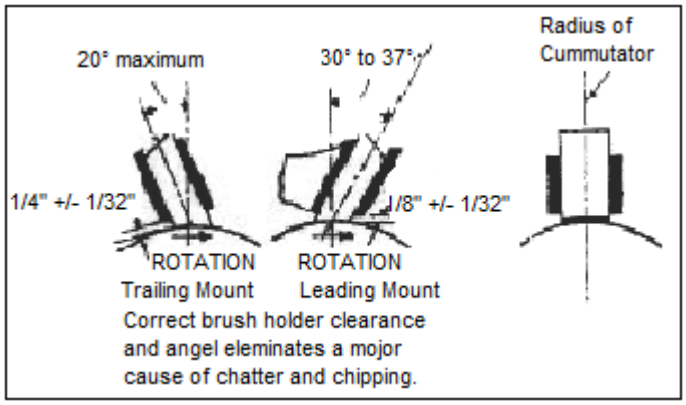


Figure 5-6. Angle of Brush Mount

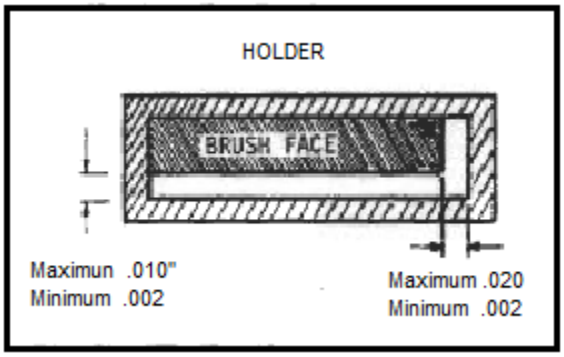


Figure 5-7. Clearance Between Brush and Brush Holder

5.2.3.2.4 Brush Seat

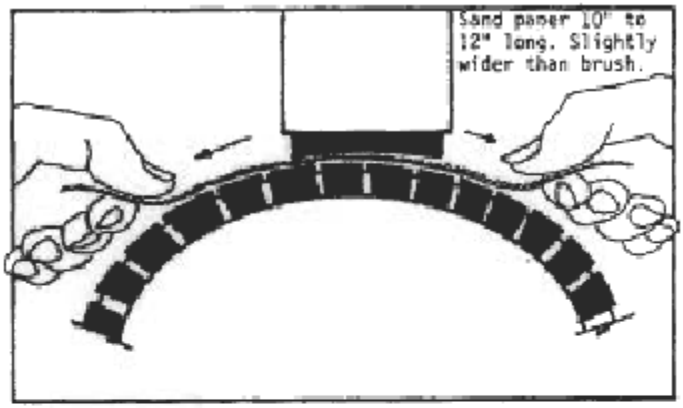
For proper seating, the face of the brush must conform to the shape of the commutator. It is not necessary or recommended that existing brushes be reseated unless the commutator has been turned. However, when new brushes are installed they require running in so that the brush face will wear to provide contact with the commutator. On large machines, only one or two brushes on each bar should be replaced at once. This allows the old brushes to carry most of the load until the new brushes are seated.

Figure 5-8 illustrates how new brushes should be seated. Use a garnet or aluminum oxide sandpaper. Never use emery paper. Emery particles are conductive and can shorten the commutator bars. This method will not make the brushes conform perfectly because of the thickness of the sandpaper results in a slightly different radius. Some run in time will be needed for best performance.

Seating stones are sometimes used to seat brushes. There are very fine grain abrasive blocks made especially for brush seating. They are used while the machine is running by holding them against the commutator in front of the brush so that abrasive particles will be carried under the brush. Some manufacturers prefer this method because it contours the brushes more accurately. However, if this method is used, the manufacturer's instructions should be followed exactly.

5.2.3.2.5 Brush Holder Alignment

Each brush holder bar should be parallel to the other holders and commutator bars. If the brushes are not parallel, the total commutation area is increased. This will result in sparking, poor commutation, and premature failure. The brushes should also be staggered as shown in Figure 5-9 so that all the commutator surface is used. This will help reduce uneven commutator wear.



1. Install new brush in holder with spring force on brush.
2. Place a strip of #0 sandpaper (not emery cloth) between the brush and commutator abrasive side to brush.
3. Pull sandpaper back and forth taking care to hold sandpaper tight against commutator until the entire face of the brush conforms to the commutator.



Like This Not Like This

4. Now place a piece of 2/0 or 3/0 sandpaper in the same position and pull in only the direction of rotation. Make several passes until the brush conforms to the commutator.

Figure 5-8. Seating a New Brush

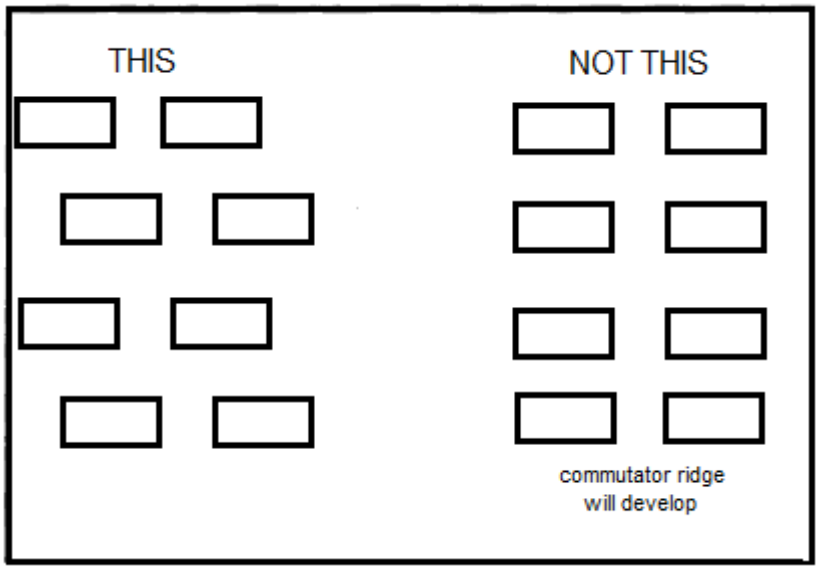


Figure 5-9. Brush Staggering

5.2.4 Brush Pressure

5.2.4.1 Determining Correct Pressure

The correct pressure between the brush and the commutator is very important. Too much pressure will cause rapid wear, while too little pressure will result in poor commutation, brush bounce, and arcing, thus often increasing brush wear more rapidly than too much pressure.

The correct brush pressure is dependent on the brush material and other factors. Brush manufacturers will usually furnish a chart showing correct pressure for their various grades of brushes. The following are typical brush pressures for broad types of brush material.

Brush Material	Pressure in PSI
Carbon	1-3/4 to 2-1/2
Graphite-Carbon	1-3/4 to 2-1/2
Carbon-Graphite	1-3/4 to 2-1/2
Graphite	1-1/4 to2
Electrographitic	2 to 3
Metal Graphite	2-1/2 to 3-1/2
Fractional HP Motor	4 to 5

5.2.4.2 Force

The force on the brush should be determined as shown in Figure 5-10. For large brushes, consider the weight of the brush in determining the total force. If the brush is on top, add the weight of brush to the reading. Take care, also, to assure that the scale is attached to the end of the spring finger where it contacts the brush. A leather loop attached to the scale is recommended for this. Note that the scale is pulled parallel to the brush. The scale should be read when the paper just slips free.

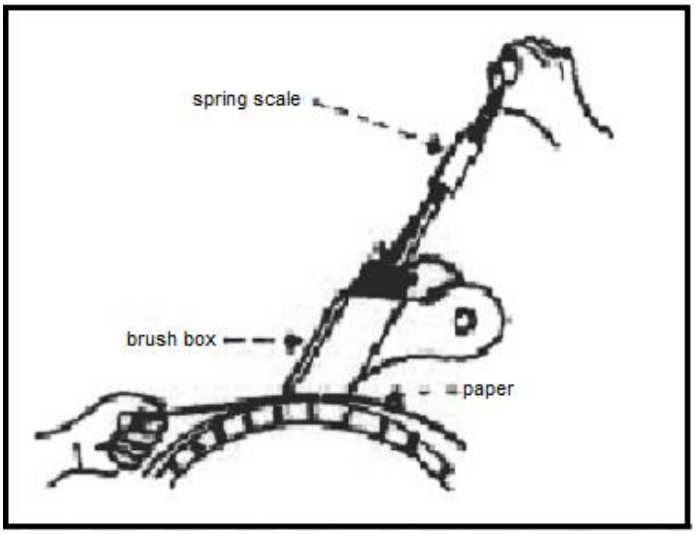


Figure 5-10. Method of Adjusting Brushes to Proper Tension

5.2.4.3 Adjusting Pressure

The scale will read in pounds or ounces. Convert this to pounds per square inch (psi) by dividing it by the face area of the brush.

Example-

A brush measures 1/2 by 1-1/2 inch. This would be an area of 0.75 square inches.

A scale reading of 4 pounds would produce 5.333 psi.

When adjusting brush pressure, always set it slightly on the high side to allow for brush wear, which will reduce pressure. Brush pressure is adjusted by various means depending on the design of the brush holder. The adjustment means will usually be obvious after careful examination. In some cases it will be necessary to replace the springs to obtain correct brush pressure.

5.2.5 Leveling Problems

5.2.5.1 General

Proper leveling is perhaps the most critical function of automatic elevator operation. Improper leveling is not only a tripping hazard, but is obvious to anyone who uses the elevator and is indicative that the elevator is receiving poor maintenance. Since most hard to solve leveling problems on DC variable voltage elevators are related to brushes and commutators, it is discussed in this subchapter.

5.2.5.2 Leveling Switch

The first impulse to correct improper leveling is to move the leveling switch. However, assuming that the switch was once set for correct leveling, this seldom corrects the problem. Unless the switch has been hit, the mechanical fastening vibrated loose, or the switch is defective, the trouble is somewhere else. The switch should be checked for sticking, bad contacts, or loose mechanical drive, but not moved until the areas discussed below have been investigated.

5.2.5.3 Slow Setting Brake

If the brake is slow in setting, the car may move after it has leveled. The direction the car moves will depend on the load in it. The brake should be carefully observed to determine if there is a perceptible delay. If so, it may be caused by one of the following:

- Improper air gap (Paragraph 5.3.4.2).
- Inadequate spring pressure (Paragraph 5.3.4.2).
- Sticking of magnetic cores due to dirt, lack of air gap, or nonmagnetic material spacers between the cores. Sometimes the spacer may be hammered too thin (Paragraph 5.3.4.2).
- Failure of economizer contacts on brake coil circuit

5.2.5.4 Soft Stop Circuit Failure

This circuit may have other names but it essentially applies a reverse current to the generator shunt field for a short time after the leveling relay drops out. Its function is to completely stop the drive motor before the brake sets. The reverse current is often supplied by a capacitor and if so, the failure of the capacitor can be the cause of leveling problems. Sticky or burned timed relay contacts may also cause the problem.

5.2.5.5 Worn Guide Shoes

Worn guide shoes can change the air gap of magnetic leveling units sufficiently to cause erratic operation. If this is causing the problem, the leveling will change with the position of the load in the car.

5.2.5.6 Improper Compounding

If the elevator is a new installation, this should be corrected by the contracting installer. If leveling becomes erratic after a short period, it may be due to generator brushes seating better and affecting compounding. To check for this, put a full load in the car and operate it in both directions, checking leveling speed. If the up leveling speed is higher than the down leveling speed, the generator is over compounded. If the up leveling speed is less than the down leveling speed it is under compounded. Refer to the control drawing to correct by increasing or decreasing resistance in the generator series field. On new installations, the contractor installing the units should make needed adjustments. If the leveling changes from cold to hot the compounding is probably being changed by changing brush to commutator resistance. This may be caused by contamination on the commutator or the wrong brush grade, improper brush pressure, etc. Refer to Paragraph 5.2.3 for information on correcting this.

5.2.5.7 Leveling Speed Change

This may be caused by the wearing in of new parts, reducing friction. If this is the case, readjust leveling speed with balanced load. The shunt field of the hoist motor can also change the speed. Check the field voltage. Low voltage may be due to a bad field strength relay. Also, check the field for a shorted winding. A shorted winding will increase both the leveling and top speed. If the voltage or resistance across one field winding is less than the others, a short is indicated and the winding should be changed.

5.2.5.8 Decreased Leveling Speed

This may be due to improper compounding, high friction in new parts, or low exciting voltage. Improper grounding on a bad generator may also cause reduced speed and erratic leveling. The wiring diagram must be carefully studied and understood to correct these problems.

5.2.5.9 Leveling on AC Elevators

Improper leveling of elevators using AC hoist motors may be caused by one of the following:

- Improper brake lining. Since these brakes have to stop the car as well as hold it, the lining is subject to wear and changes of friction characteristics. If the lining has been changed, it should be suspected and checked to be sure the correct lining material was used.
- Defective switch. A bad contact, loose mechanical drive or mounting, or sticking switch may be the problem.
- Worn guide shoes.
- Brake not setting correctly.
- Failure of two-speed control. In two speed AC machines, the high speed should drop out and the slow speed stabilize before stopping. Otherwise stopping will occur from varying speed and result in erratic leveling.

5.3 ELEVATOR BRAKE

5.3.1 General

Elevator brakes have many appearances. However, there are two basic types, alternating current (AC) brake and direct current (DC) brake. The AC brake has several disadvantages and consequently, is decreasing in use even on AC hoist machines. The availability of low cost solid state rectifiers to supply DC power for brakes makes the installation of DC brakes on AC machines practical.

The brake on the elevator drive motor shall be appropriately maintained to ensure proper operation. This includes but is not limited to residual (anti-magnetic) pads, lining and running clearances, pins and levers, springs, sleeves and guide bushings, discs and drums, brake coil and plunger, and any other part of the main machine brake system not listed above.

If any modification is made to the brake, or any part of the brake system is changed or adjusted in such a manner that the brake holding or decelerating capacity is affected, its proper function and holding capacity shall be verified as per ASME A17.1.

5.3.2 Code Requirements

The basic code requirements for elevator brakes are found in Rule 2.24.8.1 of ASME-A17.1.

NOTE that:

- The brake is applied when power is off.
- The brake must be capable of holding the car with 125% of rated load for passenger elevators and freight elevators allowed to carry employee passengers.

These rules were developed by experience and should be understood by maintenance personnel. Even a minor change in an elevator circuit can cause a violation of the rules and result in a serious accident.

5.3.3 AC Brake

The brake on an AC hoist machine serves to stop the hoist machine as well as hold it. Consequently, the AC machine brake must be of sufficient capacity to dissipate all energy from motion of the elevator. This load is reduced where a multiple speed hoist motor is used. The brake will normally be required to stop the elevator from its slowest speed. Because of the large spring required on an AC brake and characteristics of AC magnets, starts and stops with an AC brake are usually abrupt and displeasing. AC brakes usually use a dashpot type damper (either oil or air) to retard their action and reduce the abruptness. The AC magnets have the same characteristics and disadvantages of AC relays discussed in Paragraph 5.1. This includes a requirement for shading poles and need for fast action to reduce current requirements and prevent overheating. The manufacturer's instructions for maintenance and adjustment should be followed closely to assure optimum performance.

5.3.4 DC Brake

5.3.4.1 Description

Unlike the AC machine, the function of the brake on a DC machine is primarily to hold the machine after most of the stopping has been accomplished by dynamic braking through the hoist motor. Also, the characteristics of the DC magnets, like those of the DC relay, lend themselves to control action for smoother operation.

5.3.4.2 Brake Adjustments

The manufacturer's instructions for each type of brake should be followed for brake adjustment. The following will aid in understanding the manufacturer's instructions.

- The brake shoe hinge pins are on the center of the brake wheel and the entire assembly is centered over the brake wheel. The brake will not function satisfactorily if this alignment is not made.
- The brake shoe adjusting screw is used to provide equal clearance between each shoe and wheel at both the top and bottom. This may vary between .002" and .020” in the released position. But it should be uniform throughout the length of the shoe. Improper adjustment can result in brake failure to hold, rapid lining wear, and/or overheating.
- The spring force that applies the brake is adjusted by turning the nut or bolt at the end of each spring. This should be adjusted just tight enough to hold the car as required by the code. Overtightening requires more current to operate the release magnet and causes poor brake operation. If the adjustments in two previous bullets are not correct, there will be less than full contact between the brake shoe and wheel, which will require a greater spring force to set the brake. If separate adjustments are provided, care must be taken that they are adjusted equally to provide the same force on each shoe. The manufacturer's instructions will usually specify the method for equal adjustment.
- The hinge pin on the shoe should be free to turn but not be loose in the hole. Note that the plunger movement is only 1/8 to 3/16 inch, and loose pins prevent proper brake operation. Loose pin holes should be reamed and larger pins or bushings installed.
- The magnetic cores of the brake should not touch in the center because the residual magnetism will cause them to stick and delay or prevent the brake from setting. Some manufacturers use a spacer of non-magnetic material between them while others use an air gap stop adjustment
- The remaining adjustments are electrical adjustments. Electrical adjustments should be made last and only when you are satisfied with the mechanical adjustments.

5.3.5 Maintenance

5.3.5.1 Caution

Do not make any brake adjustment without taking the car out of service. Do not remove any part of the brake without first landing the counterweight. Otherwise the brake may release and cause damage and/or injury due to an overhauling load.

5.3.5.2 Cleaning

The most important aspect of brake maintenance is cleanliness, especially on the brake lining. Take special care to prevent oil or grease from dripping on the lining or drum. When pins or other moving parts are lubricated, all excess lubricant should be wiped clean. The brake drum is often adjacent to the traction sheave. In these instances, over-lubrication of ropes will not only reduce traction but will also damage the brake operation.

5.3.5.3 Lubrication

The pins on the shoes and other moving parts should be lubricated periodically according to the manufacturer's instructions. Once a year the magnetic cores should be removed, cleaned, and lubricated. The brass sleeve which houses the magnetic wires should be cleaned and replaced if worn. On horizontal cores, this sleeve can be turned 180° rather than replacing it. After cleaning, coat the wires with light oil and then wipe the oil off thoroughly with a clean soft cloth. The cores are then sprinkled with powdered molybdenum disulphide which will stick to the core without gumming. If molybdenum disulphide is not available, powdered graphite can be used. Take special care not to spill any lubricants on the brake lining or wheel. After performing this service the brake should be assembled and adjusted.

CAUTION

The counterweight must be landed before the brake is disassembled.

5.3.5.4 Brake Lining

With good care and adjustment, brake linings on DC machines can last 20 years. The lining should not be allowed to wear to where the rivets contact the wheel or the brake will lose its holding power and also damage the wheel. Always use only lining specified by the equipment manufacturer for replacement.

SECTION 6

SAFETY DEVICES AND ELEVATOR ROPE

6.1 GENERAL

All elevators except hydraulic elevators are required to have safeties and governors. Note that the governor rope is attached to the car at the release carrier and the governor is turned by the motion of the car. When the governor trips, the governor jaws grip the rope and pull it out of the release carrier, thus pulling the safety tail rope or trip rod and setting the safety. The purpose of this is to prevent the car from over speeding. Counterweights are also required to be provided with governors and safeties where the hoisting does not extend to the lowest floor of the building and the space below the hoistway is not permanently secured against access.

6.2 SAFETY DEVICES

All safety devices shall be appropriately lubricated and free from anything that might interfere with their operation (Refer to Section 3, CLEANING AND LUBRICATING).

6.3 GOVERNORS

6.3.1 Description

Two typical governors are the flyball (vertical shaft) and horizontal shaft governors. The flyball (vertical shaft) have special rope gripping jaws that grip the governor rope when the governor trips. The horizontal shaft governors have a mechanism that clamps on the rope and governor sheave, retarding the motion of both.

Governors shall be examined on a regular basis to ensure that all seals are intact and shall be hand-operated to verify all moving parts, including the rope-grip jaws and switches, move freely. As with all other elevator assemblies, the governors, governor ropes, and sheaves must be kept free of obstructions and/or contaminants that might interfere with their operation, which applies to the build-up of lubricants and/or materials in the grooves of those governors or sheaves.

6.3.2 Governor Switches

A governor overspeed switch, arranged to remove power from the driving machinery when the governor trips, is required on all counterweight governors and on car governors with type B & C safeties and rated speed over 150 fpm. These switches are set to trip at 90% of the governor trip speed for car speeds up to 500 fpm and 95% for car speeds over 500 fpm. Some governors also have speed reducing switches. When speed reducing switches are installed, the setting of the overspeed switch can be changed. See Rule 2.18.4.2. of ASME-A17.1 for correct governor switch setting.

6.3.3 Governor Trip Speeds

The trip speed of the governor is adjustable, so the setting is very important. The speed is set in the presence of an inspector by using a wheel attached to a variable speed electric drill to spin the governor and a tachometer to measure the speed. The rope is removed from the governor while this is done and the inspector seals the adjustment with a lead and wire seal. In no instance may this seal be removed by maintenance personnel. This adjustment is made at least every 5 years or more often if required by the elevator inspector. The minimum governor trip setting is 115% of rated car speed and the maximum is given in Table 2.17.3 of ASME A17.1.

The governor rope is made of iron, steel, monel, phosphor bronze, or stainless steel. They are of regular-lay construction. When the governor rope is replaced, the replacement must be the size, material, and type of construction specified on the governor marking

6.3.4 Governor Maintenance

The governor must be kept clean and properly lubricated. The rope gripping jaws and governor rope must be kept clean and free of contamination, which might inhibit rope gripping action. Governor rope should never be lubricated after installation because the lubrication would become excessive and cause the rope to slip on the governor sheave and through the gripping jaws. Therefore, the lubricant of the new governor rope must last the life of the rope.

Be careful when painting to assure that the paint does not interfere with moving parts. After completing painting, the governor must be checked for free movement and proper operation.

6.4 SAFETIES

6.4.1 General

The safety device is designed to stop the elevator car to prevent over speeding; its operation is caused by the governor. Neither safeties or governors are normally used on direct plunger hydraulic elevator. However, they are not prohibited.

6.4.2 Types and Applications

6.4.2.1 Type A Safety

Referred to as an instantaneous safety, it applies a rapidly increasing pressure to the rails and stops the car in a very short distance. These can only be used on elevators having a rated speed of 150 fpm or less.

6.4.2.2 Type B Safety

This type of safety applies a limited pressure to the rails to provide uniform retarding forces. Continuous tension in the governor rope may or may not be required. Minimum and maximum stopping distances are specified in Table 2.17.3 of ASME A17.1.

6.4.2.3 Type C Safety

A type C safety is essentially a type A safety with an oil buffer between the car frame and safety plank. The stopping distance is the effective stroke of the buffer. This type may be used for car speeds up to 500 fpm provided other specific code requirements for this safety are met.

6.4.3 Maintenance

Maintenance of safeties is very important because if a safety fails to operate properly, serious damage to equipment and personnel injury will result. All moving parts of the safeties must be cleaned and properly lubricated.

6.4.4 Lubrication

Care must be exercised to prevent over-lubrication. Note that safety bearing surfaces are required to be made of corrosion resistant materials. The tail rope (rope used to connect the safety to the governor rope) is also required to be of corrosion resistant material. Improper type or excess lubricants on the guide rails will reduce the holding power of safeties. Therefore, only lubricants approved by the manufacturer of the safety may be used. If rail lubricants are required, attach a metal plate to the crosshead stating the following: *Consult Manufacturer of the safety for characteristics of the rail lubricant to be used*. If rail lubricants are not required, the plate must read: *Rail Lubricants are not required*.

Rails on which lubricants are used must be thoroughly cleaned at least once a year with a safe solvent to remove excess lubricants, lint, and dirt accumulation. Also rust preventive compounds such as paint, graphite oil, etc., may not be used on the rail.

6.5 BUFFERS

6.5.1 Function

The buffer prevents damage to the equipment or serious discomfort to passengers if a control malfunction causes the car or counterweight to run into the pit at rated speed. The buffer serves to provide a rate of deceleration, which will stop the car of counterweight without damage.

6.5.2 Types

Spring Buffers:

Typical coil spring buffers are shown in Figure 6-1. This type of buffer is allowed where the rated speed does not exceed 200 fpm.

6.5.2.1 Oil Buffers

6.5.2.1.1 Description

Typical oil buffers are shown in Figure 6-2. When the plunger of an oil buffer is depressed, oil under the plunger is forced to a second chamber through an orifice. Thus the speed which the plunger can be depressed by a given force is dependent on the orifice site and the viscosity (thickness or weight) of the oil. Therefore, the proper oil must be used in the buffer. Oil buffers also have a spring to return the plunger to the fully extended position after the load is removed.

6.5.2.1.2 Marking

The marking plate shall conform with Rule 2.22.4.11 of ASME A17-1.

6.5.3 Maintenance

6.5.3.1 Spring Buffers

Maintenance of spring buffers consists of checking their mounting and keeping foreign matter out of the coils. Also, if a pit is subject to water, severe corrosion can damage the spring.

6.5.3.2 Oil Buffers

Oil buffers require a monthly check of the oil level and condition of the plunger. The plunger return must be tested annually by fully depressing it and determining that it will return to its fully extended position within 90 seconds. Every 5 years, test the buffer by running the car with rated load (or counterweight with no load in car) into the buffer at rated speeds. A metal tag must be attached to the buffer showing the date of the test and person or firm that conducted the test.



Figure 6-1. Typical Coil Spring Type Buffer (used only where car speed is less than 200 fpm)



Oil Buffer Marking Plate
Specified in Rule 201-4K



Figure 6-2. Typical Oil Buffer and Marking Plate

6.6 ELEVATOR ROPES

6.6.1 General

6.6.1.1 Rope Strength

Correct wire rope in good condition is essential to safe efficient operation of the traction elevator. Wire ropes is the preferred terminology, but they are sometimes called cables. This comes from their strength comparison to the larger fiber ropes or cables of 16 to 24 inches in diameter which were used on ships. The first wire ropes were made in Germany in the 1830s, and shortly thereafter wire ropes were made in the United States. Typical strength of two types of elevator wire ropes is shown in Table 6-1 and Table 6-2.

6.6.1.2 Rope Components

The components of wire rope are shown in Figure 6-3. Note that each strand is made up of a number of wires twisted (layed) around a center wire. All the wires in the strand may or may not be the same size. In some rope, the core is a smaller wire rope, but in elevator rope, the core is always a fiber. This provides a reserve for lubrication. The fiber core ropes are specially made to close tolerances for this purpose. They may be natural fiber or synthetic, but natural fibers are usually used. Figure 6-4 illustrates a rope lay. A lay is the distance or length of rope required for one strand to make a complete circle around the rope.

Table 6-1. Typical Strength of Elevator Wire Ropes

ELEVATOR WIRE ROPES, 6 X19 CLASS, REGULAR LAY, FIBER CORE					
		Breaking Strength, lb			
Rope Diam. In.	Approx. Weight per ft. lb.	Extra High Strength Traction Steel	High Rise Special Traction Steel	Traction Steel	Iron
1/4	.10	5,200	---	3,600	2,200
5/46	.16	8,100	---	5,600	3,200
3/8	.23	11,600	---	8,200	5,000
7/16	.31	15,700	---	11,000	6,400
1/2	.40	20,400	---	14,500	8,400
9/16	.51	25,700	---	18,500	10,600
5/8	.63	31,600	---	23,000	12,800
11/16	.76	38,200	30,000	27,000	---
3/4	.90	45,200	---	32,000	18,200
13/16	1.06	52,900	46,000	37,000	---
7/8	1.23	61,200	---	42,000	24,800
15/16	1.41	70,000	60,000	48,000	---
1	1.60	79,500	---	54,000	32,000

Table 6-2. Typical Strength of Elevator Wire Ropes (Continued)

ELEVATOR WIRE ROPES, 8 X19 CLASS, REGULAR LAY, FIBER CORE					
		Breaking Strength, lb			
Rope Diam. In.	Approx. Weight per ft. lb.	Extra High Strength Traction Steel	High Rise Special Traction Steel	Traction Steel	Iron
1/4	.09	4,500	---	3,600	1,800
5/46	.14	6,900	---	5,600	2,900
3/8	.20	9,900	---	8,200	4,200
7/16	.28	13,500	---	11,000	5,600
1/2	.36	17,500	---	14,500	7,200
9/16	.46	22,100	---	18,500	9,200
5/8	.57	28,200	---	23,000	11,200
11/16	.69	32,800	30,000	27,000	13400
3/4	.82	38,900	---	32,000	16,000
13/16	.96	46,000	46,000	37,000	18600
7/8	1.11	52,600	---	42,000	21,400
15/16	1.27	60,000	60,000	48,000	24600
1	1.45	68,400	---	54,000	28,000
1 1/16	1.64	77,000		61,000	31,400

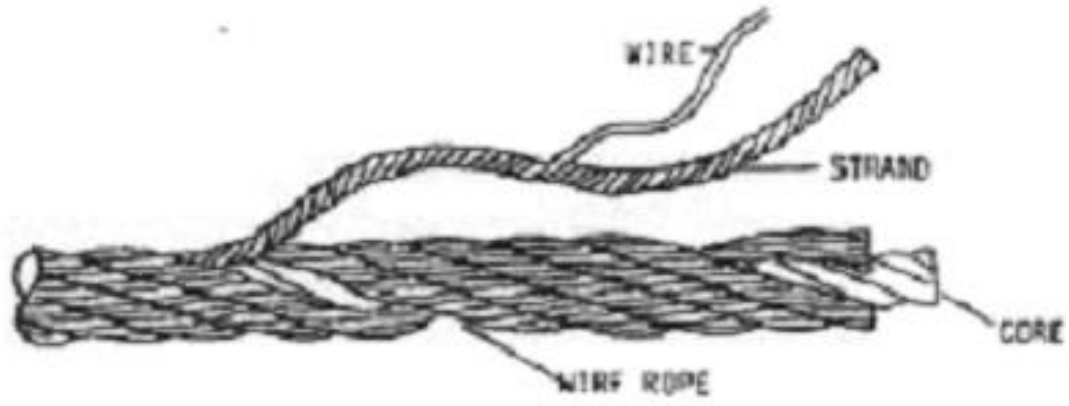


Figure 6-3. Components of Wire Rope

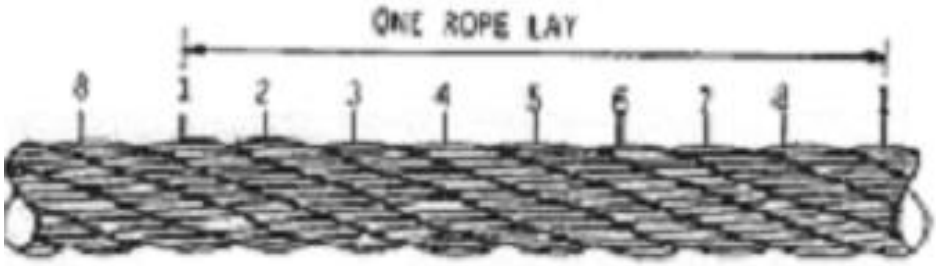


Figure 6-4. Components of Wire Rope

6.6.2 Wire Rope Construction

6.6.2.1 Characteristics

6.6.2.1.1 General

The ways that wire ropes are made affect the characteristics of the ropes.

6.6.2.1.2 Type of Lay

In regular lay, the exposed wires are parallel to the rope. In lang lay rope, the wire in the strand and the strand are layed in the same direction. This results in longer length of exposed wire and thus greater bearing surface between the rope on sheave than the regular lay rope. The arc of bending of the wires around a sheave is also greater for a lang lay rope, resulting in less bending stress and fatigue. However, when a wire breaks in a non-preformed lang lay rope, it is much more likely to get wrapped around the rope and cause failure of other wires, since the length of broken wire is longer than in a regular lay.

6.6.2.1.3 Preformed Rope

In preformed rope, each wire and strand is permanently formed to the shape it will have in the rope before the strands are layed around the core to make a rope. Therefore, they tend to stay together. The wire and strand in Figure 6-3 are retaining their preformed shapes.

6.6.2.1.4 Nonpreformed Rope

In nonpreformed rope, straight wire and strands are twisted and held in place by the seizing at the end of the rope. If the seizing is cut on a nonpreformed rope, each wire will tend to straighten out and the rope will come apart. Nonpreformed ropes are subject to both the stress of the rope shaping and the stress of the load. Also, when a wire breaks, the broken ends tend to straighten out and often damage other wires.

6.6.2.1.5 Strand Construction

Figure 6-5 shows four strand construction common to elevator rope. Note that the wires are not all the same size. Different sizes are used to fill the voids as much as possible. The 8 x 19 seal is very common in elevator rope.

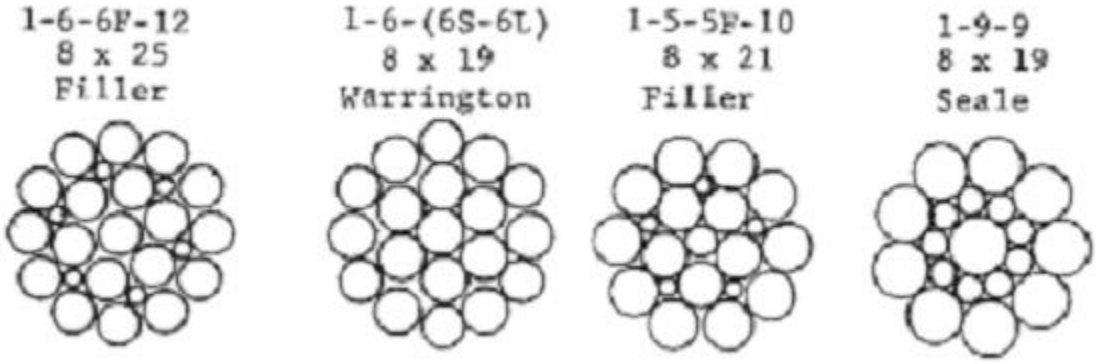


Figure 6-5
Strand Construction

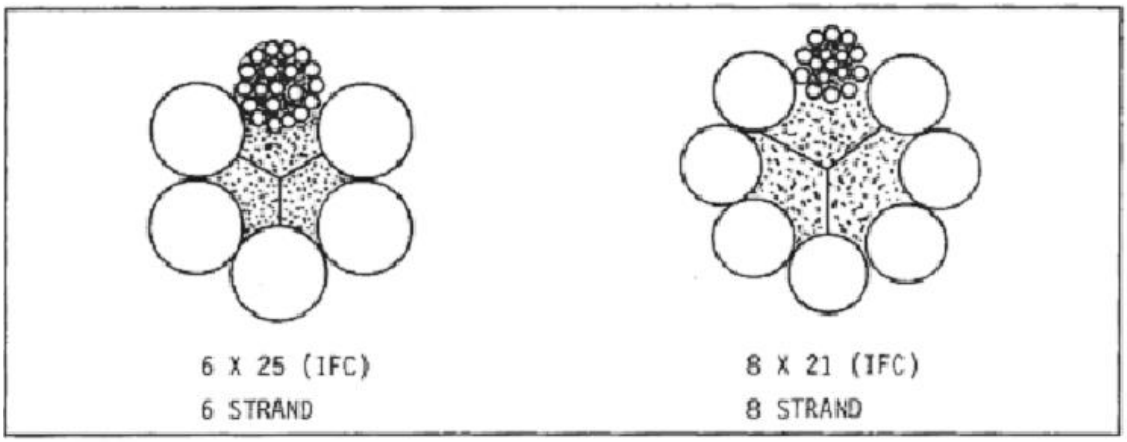


Figure 6-6. Wire Rope Cross Section

6.6.2.1.6 Rope Cross Section

6.6.2.1.6.1 Wire Rope

The cross section of two common types used for elevator hoist rope are shown in Figure 6-6. Note that the eight-strand rope has a larger fiber center. This larger fiber center results in the eight strand rope being more flexible and conforming better to the sheave grooves. It also has more strands in contact with the sheave.

6.6.2.1.6.2 Tiller Rope

Figure 6-7 shows Tiller rope construction. This is a very flexible type of construction. However, it should not be used for hoist or governor rope. The only exception is on very old governors that were specifically designed for Tiller rope.

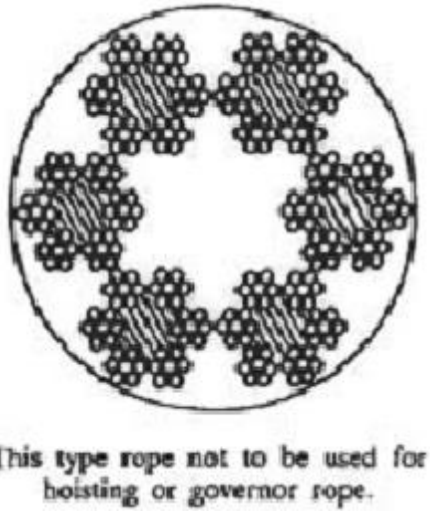


Figure 6-7. Tiller Rope Construction

6.6.2.2 Code Certification

The code (ASME A17.1) required that only iron (low carbon steel) or steel wire rope specifically constructed for elevator be used. All hoist ropes must be from the same reel. A certificate certifying that the rope meets the elevator code should be furnished.

6.6.3 Fastening

The fastening of the rope is the weakest point in the rope. The hoisting rope sockets are required to develop at least 80 percent of the ultimate breaking strength of the rope. The design of the tapered socket is specified in Rule 2.20.9.7 of ASME A17.1 This also specifies the method of socketing using Babbit Metal. The procedures specified in the code should be followed and the complete socket inspected by a qualified elevator inspector.

6.6.4 Maintenance and Handling

6.6.4.1 Storage

Wire rope should not be stored for extended periods (several years) or the lubricant will dry out and the fiber deteriorate. Rope in storage should be turned periodically to prevent the lubricant from collecting in one part.

6.6.4.2 Handling

Rope should be handled in a manner to prevent kinking, untwisting, or exposing to corrosive materials.

6.6.4.3 Hoist Rope Tension

Hoist rope tension should be kept equal so that the load is equally distributed. If the ropes are not under equal tension, the loose ropes and their sheave grooves will show excessive wear. If the sheave grooves are of unequal depth, creeping, excess wear, and loss of traction will result.

6.6.4.4 Lubricant

6.6.4.4.1 Reasons for Lubricating

When elevator ropes are manufactured, each wire is coated with a special lubricant and the fiber center is impregnated with the lubricant. This initial lubricant will not last the life of the rope, however. When the rope is in operation, each wire is in rubbing contact with several others. If inadequate lubricant is present, an abrasive action occurs between each wire. This results in a fine powder which quickly oxidizes and forms a red dust between the strands. This red dust acts as an abrasive and accelerates the wear. Since this wear is in the interior, broken wires may not be observable even though the strength of the rope is severely reduced. When this happens, the hoist ropes will have to be replaced.

6.6.4.4.2 Procedure

See Paragraph 3.2.4 for procedure for lubrication of ropes. Note again that governor ropes should never be lubricated after installation. The lubrication may interfere with the holding action of the governor jaws. However, governor ropes should be kept clean and free of dirt build-up.

6.6.5 Rope Data

6.6.5.1 Hoist Ropes

6.6.5.1.1 Data Plate

The rope data plate should be attached to the car crosshead, and show the required information (Rule 2.20.2 of ASME A17.1):

- The number of ropes.
- The diameter in inches.
- The manufacturers rated breaking strength per rope in pounds.

6.6.5.1.2 Data Tag

A metal data tag must also be attached to one of the ropes showing the following:

- The diameter in inches.
- The manufacturers rated breaking strength.
- The grade of material used.
- The month and year the ropes were installed.
- Whether nonpreformed or preformed.

6.6.5.2 Governor Rope

A metal plate should be attached to each speed governor showing-

- The speed in FPM at which the governor is set and sealed to trip the rope gripping jaws.
- The size, material, and construction of the governor rope on which the governor jaws were designed to operate.

6.6.6 Wire Rope Inspection

The wire rope inspection procedure and criteria are contained in American National Standard Practice for the Inspection of Elevators, Escalators and Moving Walks, ASME A17.2.

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SECTION 7

HOISTWAYS, DOORS, AND GATES

7.1 HOISTWAYS

7.1.1 Fire Rating

Elevator hoistway enclosures are required to have a fire resistance rating in compliance with the building codes and Rule 2.1.1.1 of ASME A17-1. Note that each hoistway enclosure is not tested, but the type of material and construction should be verified by the architect as having been tested and meeting the fire resistance requirements. Damage to hoistway walls must be repaired promptly to maintain the integrity of the enclosure.

7.1.2 Windows

Windows in the hoistway wall are prohibited. Windows in hoistway walls on the interior, such as around stairs, which were common in older buildings, do not meet current fire rating requirements. Windows in exterior walls create a hazard for firemen in a fire or other emergency. Where windows exist, a repair project to close them must be initiated. Meanwhile, immediately install a warning sign identifying any window leading into an elevator hoistway.

7.1.3 Other Equipment

There is a temptation to install utility lines, communication equipment, etc., in elevator hoistway. This is in violation of the code and is not permitted in postal buildings under any circumstances. The main power feeders for the elevator must be installed outside the hoistway. The only equipment allowed in elevator hoistways is equipment used directly in connection with the elevator.

7.1.4 Vents

Vents are required for elevator hoistways in elevators serving more than three floors, for venting smoke and hot gas to the outside in case of fire. Since hoistway doors are not air tight, a relatively small fire on one floor can quickly fill an elevator hoistway with smoke and hot gas, endangering the lives of elevator passengers.

The vents must be permanently open to the outside and the area of the vent be 3-1/2 percent of the hoistway area but not less than 3 square feet per elevator. A hinged damper which will open under a very small pressure is considered a permanent opening. Hoistways are often vented into the machine room through steel gratings. The machine room is then vented to the outside. Care must be taken that these vents are not blocked and that they remain open at all times.

7.2 DOORS AND GATES

7.2.1 General

Many injuries are caused by gates and hoistway doors which are operating improperly or were installed incorrectly. Therefore, proper adjustment and maintenance is essential for safe elevator operation. Ensure hardware is tight, not excessively worn, and of the proper grade in critical areas such as the counterweight assembly.

7.2.2 Fire Resistance

Hoistway doors are required to have a fire resistance rating, as determined by test in accordance with current ASME standards. The doors must bear a label certifying their fire rating.

7.2.3 Interlocks

Hoistway doors are equipped with interlocks to prevent operation of the car until the hoistway door is fully closed and locked. The interlock prevents opening of the hoistway door from the landing side, unless the car is within the landing zone. The operation of an interlock is shown in Figure 7-1. Note that the pawl will lock before the shorting block completes the electric circuit that allows the car to operate. Car gates are not required to have an interlock, but must have an electric contact to prevent operation of the car until the gate is closed. The hoistway door interlocks and car gate contacts must never be jumped to allow operation of the car with the door open.

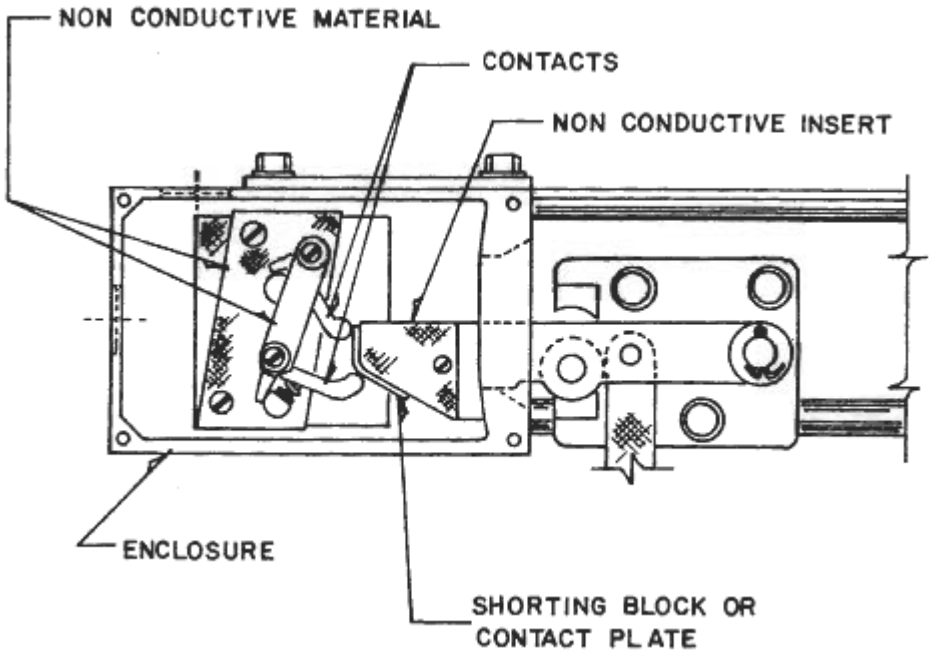


Figure 7-1. Hoistway Door Interlock

7.2.4 Closed Position of Doors

7.2.4.1 Car Doors

Car doors or gates equipped with electric contacts are considered to be in the closed position when the leading edge of the door is within 2 inches of the jam, sill, or other door panel.

7.2.4.2 Hoistway Door (Closed Position)

7.2.4.2.1 Vertically-sliding Bi-parting Counterbalance Doors

When the rigid members of the panels are 3/4 inch apart, they are considered to be in closed position. A fire-resistant, non-shearing, non-crushing astragal is required on the upper panel to close the distance between the rigid door member of the panels when in contact with stops.

7.2.4.2.2 Horizontal Sliding or Swinging or Vertically Sliding Counterweight Doors

These doors are considered to be in closed position when the leading edge is within 3/8 inch of the nearest face of the jam or sill or when the panels of horizontal sliding doors are within 3/8 inch of each other.

Horizontal Sliding Doors

These may be considered closed when they are 4 inches from jam, sill, or opposite panel, if the following conditions are met-

- The car is operated from the inside only.
- The doors are equipped with door closers that will continue to close the doors to within 3/8 inch and lock them.
- The door is equipped with interlock.

7.2.5 Door Operators

Most door operator motors use direct current, and the door closing speed can be adjusted by adjustable resistors in the motor circuit. The speed of horizontally sliding automatic doors should be adjusted so that the kinetic energy of the door is less than 7-foot pounds or 2-1/2 foot-pounds if a door reopening device is not used. Also the force to prevent closing of the door from rest must be less than 30 pounds. The door manufacturer's instruction should be referenced to determine correct means of measuring energy. A simple spring scale may be used to measure the closing force from rest.

7.3 SPECIAL PROVISIONS

7.3.1 Automatic Operation

Doors and gates which are automatic closing (or door and gates closed by momentary pressure button) are required to be equipped with reopening devices. The safety edge runs the entire length of the door. Contact with either door panel safety edge will cause the door to stop and reopen. These are often installed with photoelectric devices that will cause the door to remain open if the light beam is obstructed.

SECTION 8

HYDRAULIC ELEVATORS

8.1 HYDRAULIC ELEVATORS

8.1.1 General

A typical direct plunger electro-hydraulic elevator is generally referred to simply as a hydraulic elevator and an electric traction elevator simply as an electric elevator. Note that hydraulic elevators are not normally equipped with safeties, governors, or counterweights. They are also less expensive to install and maintain. However, they are more limited in speed and operating heights than electric elevators. Figure 8-1 shows a typical hydraulic elevator machinery set up.

8.1.2 Gland Packing and Seals

8.1.2.1 Inspection and Maintenance

Valves and cylinders utilizing packing glands and seals must be examined and maintained in order to avoid excessive loss of fluid. Oil leakage from the cylinder packing gland must be removed when it reaches five (5) gallons maximum. The container used for its collection and removal must be covered in such a way to prevent spillage or overflow.

8.1.2.2 Oil Usage Record

In those systems where visible inspection of the cylinder and/or piping is not possible, a written record shall be maintained indicating the quantity of fluid added to the system and collected from the system's overflow containers. When the quantity of fluid added does not match the quantity of fluid collected from the containers, the system shall be tested as per ASME A17.1.

8.2 HYDRAULIC JACK

8.2.1 Description

Details A & B in Figure 8-2 show the internal details of the jack. The stop ring on the end of the plunger is necessary to prevent the car from striking the overhead in event of a control malfunction. Detail B (Figure 8-2) shows the lower portions of the jack including the safety bulkhead and dished head.

8.2.2 Problems with Jacks

8.2.2.1 Damage

The jack is often damaged by electrolysis. Most jacks are wrapped with protective coating to prevent corrosion. However, soil shift or mechanical damage often punctures this coating and results in electrolysis weakening the cylinder. This can result in cylinder failure and fall of the car. There have been cases where the entire cylinder head broke off and resulted in serious personal injury. The code requirement for the safety bulkhead and dished cylinder head help reduce this danger.

8.2.2.2 Leaking Oil

Impending failure of this nature is usually signaled by a slow loss of oil. The test specified is designed to detect small pinhole leaks which signal impending failure. A log of oil added to the system will also help identify damaged, leaking cylinders. When it is determined that an elevator jack is leaking the elevator should be shut down and not used until repairs have been made.

8.2.3 Cylinder Corrosion Protection And Monitoring

8.2.3.1 Corrosion Protection Monitoring

When cylinder corrosion protection monitoring is required, the system used to accomplish that monitoring shall be maintained. If the means used to monitor the corrosion protection indicates the protection has been lost or compromised, then the corrosion protection shall either be repaired or replaced.

8.2.3.2 Single Bottom Cylinder Design Elevators

All single bottom cylinder design elevators must be taken out of service at the first sign of a cylinder leak. A cylinder leak indicates cylinder failure and impending cylinder collapse which can be catastrophic since the elevator is fully supported by the hydraulic piston, and there in no structural framework or load bearing shaft walls. Code changes in 1972 require double bottom cylinder design for all new installations. If a single bottom cylinder design elevators is found, alert the Facilities group so it can be scheduled to be updated to current code requirements.

8.2.3.3 Anti-creep and Low Oil Protection

The anti-creep function and low oil protection shall be maintained in accordance with the manufacturer's specification and the applicable code.

8.2.3.4 Packing

Detail A of Figure 8-2 shows one type of packing between the plunger and top of the cylinder. Some leakage will occur and a means should be provided to collect the leakage to prevent it from creating a hazard in the pit. The leakage shall also be measured to determine make-up requirements and help detect underground jack leaks. This oil that has been exposed to dirt and dust should not be poured back into the tank. However, some jacks have a specially designed chamber to collect and return seal leakage directly to the tank without contamination. The most common causes of excessive leakage are (a) worn seal or packing, (b) damage to the surface of the plunger, and (c) misalignment of the plunger and cylinder with the car.

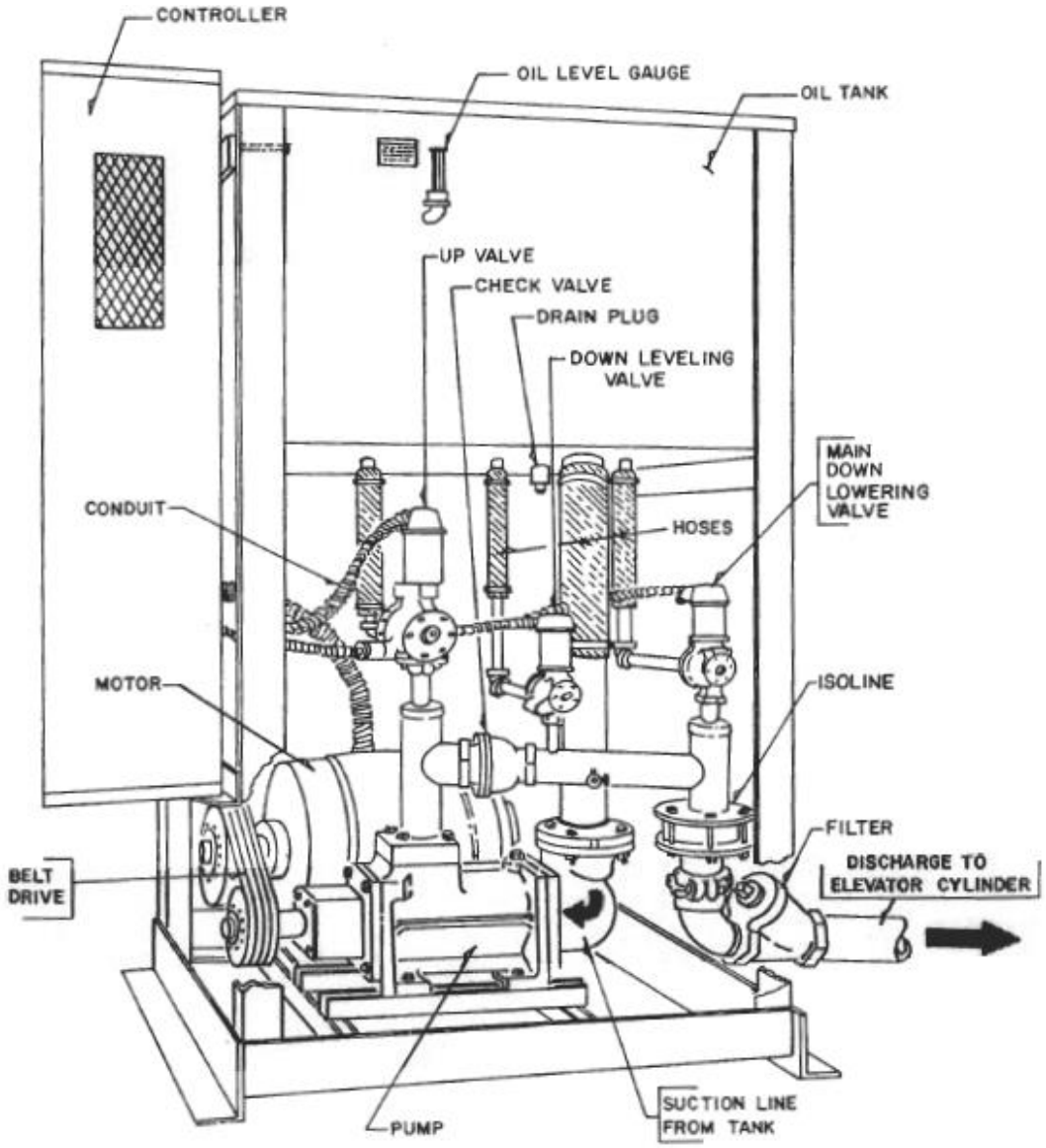


Figure 8-1. Hydraulic Elevator Machinery (from Otis Elevator Owners Information)

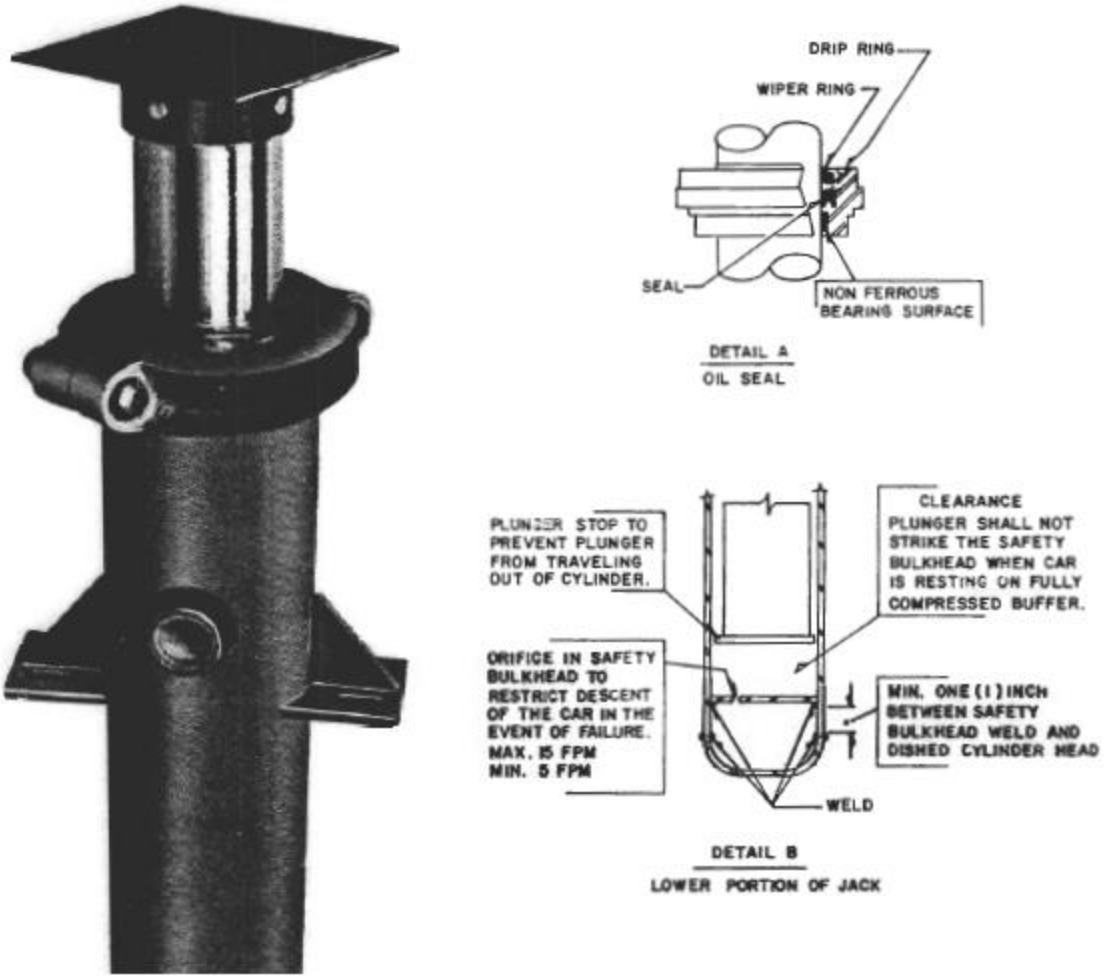


Figure 8-2. Hydraulic Jack

8.2.3.5 Plunger

The plate on top of the plunger is required to be fastened to the car. If the plunger is not fastened to the car, the plunger could go down and leave the car stuck in the hoistway to fall when it is jarred loose.

8.3 PUMPS

Many types of pumps have been used on elevator equipment. However, the gear and the small pump are the most common in use today. Since pump pulsations can be transmitted to the car, pump design and piping arrangement to dampen pulsations are very important. Most elevator pumps will discharge into a small pressure vessel (silencer) that dampens pump pulsations in the oil. Some older pumps have packing glands that require periodic tightening to prevent excessive leakage. Internal bearings in oil hydraulic pumps are usually sleeve type and will show very little wear even after years of operation unless dirt has gotten into the system. The internal parts of hydraulic pumps are precision machined for maximum efficiency and contaminated or improper oil will cause serious damage. Therefore, returning dirty oil to the storage tank or leaving the cover off the tank is prohibited.

8.4 CONTROL VALVES

Modern hydraulic elevators use electric solenoid operated valves. Figure 8-1 identifies three such valves- (1) up valve, (2) down leveling valve, (3) mail down lowering valve. Some elevators have all of the valves built into one valve body as a unit.

The functions of typical hydraulic elevator valves are as follows.

- Up Valve (sometimes called bypass valve) - This valve serves to help start the elevator smoothly in the up direction and reduce the starting torque on the motor. When the pump first starts, this valve is open, allowing some of the pump output to by-pass returning to the tank. This valve then slowly closes, directing the full output of the pump to the jack.
- Check Valve - This prevents the pressure in the jack from forcing the oil back through the pump when it is stopped.
- Main Down Lowering Valve - This valve opens to return oil to the tank and lowers the elevator.
- Down Leveling Valve - After the main down lowering valve closes, this smaller valve remains open to level the car at a slower speed.
- Hand Lowering Valve - This is for manually lowering the car in case of a power failure or equipment malfunction. This is sometimes a needle valve adjustment on the lowering valve.
- Pump Relief Valve - This is a safety valve to prevent over pressure in the system. It should be open and by-pass into the tank at a pressure not greater than 125 percent of the working pressure. It should also bypass the full output of the pump without the pressure raising more than 20% above the pressure at which it first opened. The working pressure is the pressure required to raise the elevator with rated load in the car at rated speed near the top of the hoistway. The relief valve adjustment shall be routinely inspected to ensure the valve seal is intact. If the seal has been disturbed or is not intact, the ASME A17.1 specified check shall be conducted.

8.5 PIPING

8.5.1 Pressure Rating

Piping is the often overlooked weak link in a hydraulic elevator. Schedule 40 steel pipe, which is often used in building domestic water systems, cannot be used on hydraulic elevators where the working pressure is more than 250 psig. Since most hydraulic elevators have a working pressure greater than 250 psig, schedule 40 piping should not be used.

An installer who needs only one or two fittings may be tempted to use what is available from the plumbers to get the job going. Therefore, pipe and fittings should be checked to assure that they are proper, and galvanized pipe should be suspect. High-pressure fittings are much heavier forged fittings and usually have the pressure stamped on them. Also, schedule 40 pipe and fittings are often galvanized, while high-pressure pipe and fittings are seldom galvanized.

8.5.2 Support

All piping must be provided with supports to prevent extra load (stress) on fittings, valve and tank connections, as a result of the weight of the pipe.

8.5.3 Flexible Hydraulic Hose

Flexible hoses are prohibited in the hoistway or through any wall. Where such hoses are used in machinery space, their bursting strength must be ten times the working pressure. The hose and fitting must be permanently marked, showing SAE hose type and required replacement date (not more than 6 years from the installation date). Hydrostatic test is required every 3 years and replacement every 6 years.

If it is discovered that there is no record of the original installation date, the hose assembly shall be replaced immediately and the installation date of the new hose appropriately recorded. All replacements shall conform to ASME A17.1.

SECTION 9
ESCALATORS
(RESERVED)